Dense Retrieval Is Degenerate Associative Memory: A Reduction and a Regime Taxonomy for Agent Long-Term Memory

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Abstract

Agent long-term memory is almost universally retrieved by top-k embedding similarity. Associative-memory theory offers an alternative—recall as convergence to an attractor in an energy landscape—but it has never been evaluated as the *persistent memory layer* of an agent under matched conditions. We first show why the two literatures have not met: at the standard operating point they are the same operator. Proposition 1 establishes that dense retrieval is exactly the (T=1, $\beta \to \infty$, b=0, R=I) corner of the modern Hopfield update — exactly at k=1, and for $k > 1$ only when the shortlist is read off the cue rather than off the converged state, which is a distinction we measure rather than assume. It is verified numerically on 1000/1000 queries on synthetic vectors and 838/838 on real sentence embeddings. The industry baseline is therefore a configuration of our own code rather than a separate system, which removes implementation advantage by construction and converts an unanswerable question into a sharp one: *which departures from that corner pay for themselves, and where?*

We answer it across three modalities and twelve corpora with matched encoders, group-disjoint hyperparameter selection, and cluster-bootstrap intervals throughout: 5,847 held-out image queries, 1,977 LoCoMo and 479 LongMemEval questions, 5,435 questions over five retrieval corpora, and 63,763 temporal-association probes across three corpora. The answer is negative for every mechanism that changes only *how an answer is read off* a fixed landscape, and the clearest evidence for that is the collapse of our own strongest positive result. Iterative descent beat cosine by $+0.014$ pooled $[+0.006, +0.021]$ on occluded images, rising to $+0.021$ at 70% occlusion. That number reproduces exactly and means much less than we claimed: the metric is ten-way same-class accuracy, the cue's source image is never in the bank, and at the selected β the first descent step averages 12.7 patterns out of 400 with half the blend mass on the cue's own class — an implicit class vote. Scored against the matched baseline that makes explicit, a soft k-nearest-neighbour class vote with no energy, no iteration and no attractor, descent is *worse* at every occlusion level (-0.036 pooled, [-0.043, -0.028]). Building the experiment our premise actually describes — occluded copies of images that *are* stored, scored by identity rather than class — leaves descent at -0.003 [-0.008, +0.000] against cosine. Descent loses monotonically on conversational question→evidence retrieval as well: every increment costs accuracy, from 0.336 to 0.252 Coverage@5 across an interpolation sweep whose endpoint provably reproduces cosine.

One premise explains every loss. Associative completion requires the cue to be a degraded instance of a stored pattern, and question answering violates it. We nominated the discriminating tests against ourselves and ran all three: masked spans of stored *text* satisfy

the premise and give no β that beats cosine in 45 swept cells; a cue that is a degraded copy of *two* stored images at once loses -0.245; and occluded copies of single stored images, the premise's best case, are the null above. The premise therefore predicts every loss we measure and retains *no* confirming instance — we report it as necessary and not sufficient, and withdraw the image result as evidence for an associative mechanism.

We further show that β—the only dial with leverage—is influential only inside a narrow, encoder-specific band ($\approx[32, 64]$ on the vision backbone we measure it on), above which descent degenerates into cosine — at β=512 it returns cosine's own memory on 99.4% of queries. A comparison run above that band compares cosine against itself, which we argue is a live hazard for this literature rather than a detail of ours. Of eight mechanisms tested beyond the reduction corner, exactly one survives: a nonparametric predictor over the agent's own observed transitions beats cosine on one-hop temporal association, reproduced in four measurements over three corpora. On LoCoMo, against a pooled bank of 5,882 turns and over 5,233 probes in 272 sessions, it raises Recall@1 from 0.064 to 0.107 ($+0.043$, $[+0.036, +0.050]$, $p < 10^{-4}$; +67% relative) and Recall@5 from 0.174 to 0.251; the same driver, run unmodified on MSC's 30,320-turn bank, gives $+0.016$ $[+0.014, +0.018]$ over 27,815 probes. The multi-hop extension does not survive: at two hops the effect is -0.002 $[-0.008, +0.004]$ on LoCoMo and $+0.005$ $[+0.003, +0.006]$ on MSC, a factor of 18 below the $+0.185$ two-hop result of our own that we withdraw. Multi-step descent, residual peeling, repulsive parallel descent, vote-count readout, multi-scale β, per-memory β, retrieval-margin calibration, and coupling the predictor into the energy functional all fail, each with a diagnosed mechanism rather than a bare null.

One departure from the corner does pay, and we found it by running the six corpora that LoCoMo alone could not speak for. Folding K_0 retrieved memories back into the cue and retrieving again beats dense retrieval by $+0.023$ $[+0.010, +0.037]$ on HotpotQA bridge questions and $+0.171$ $[+0.118, +0.219]$ on 2Wiki inference questions, survives a disjoint sample, a second encoder and two bank sizes, and beats a random-mode control by $+0.13$ to $+0.39$ in 18 of 18 cells — so the gain is in *which* memories enter the cue, not in expanding it. On comparison questions in the same corpora, where both documents are reachable from the question directly, selection declines to expand in 10 of 10 folds. What predicts the gain is hop type, not target count: our own pre-registered hypothesis that more targets should help more comes out *reversed* on four corpora and we withdraw it. The result is sharply bounded — it is query expansion rather than attractor dynamics, it gives up its entire gain when its configuration is transferred across corpora, and on both conversational-memory corpora it is null, where a lexical hybrid gains $+0.156$ $[+0.125, +0.189]$.

Finally, we report the measured magnitude of seven protocol errors we committed and corrected. Each would have yielded a publishable but false result: a non-dimensionless hyperparameter (0.016 vs. 0.277 after normalisation), a pseudo-replicated bootstrap (p=0.0005 vs. p=0.076), test-set hyperparameter selection ($+0.042$ vs. $+0.004$), asymmetric candidate exclusion (an exact 0.000 that read as a finding), an unmatched step size (-0.115 vs. -0.062), and a significant result whose script was never committed and does not reproduce ($+0.049$, p=0.032 vs. $+0.024$, p=0.24). Two of the seven were false *negatives*, which proved materially harder to find. We release all code, the regime taxonomy, and an append-only experiment ledger in which every entry above appears with its date, its question, and the entries it overturned.

1 Introduction

A large language model is a function from text to text with no state between calls. An *agent* is what results when that function is asked to operate over days or months, and the forgetting becomes the binding constraint. The standard remedy is an external store: write experience to it, and on each turn retrieve a few items to place back in context. The retrieval step is the subject of this paper.

Across the agent-memory literature—MemGPT (Packer et al., 2023), MemoryBank (Zhong et al., 2024), A-MEM (Xu et al., 2025), HippoRAG (Gutiérrez et al., 2024)—the sophistication is concentrated almost entirely in *what gets written and when it is evicted*. The read operator itself has been a frozen cosine nearest-neighbour lookup for a decade. Three objections to it are standard. It gets one shot at an ambiguous cue; it weights a memory recalled four hundred times identically to one glimpsed once, patched by hand-tuned recency and usage heuristics bolted to the side of a similarity score; and it can only return what is *similar*, which is structurally the wrong relation for recall by association—lightning does not resemble thunder.

A separate literature has studied exactly this operator for forty years. Associative memory treats recall as a dynamical process: a cue is released into an energy landscape whose minima are the stored patterns, and it descends until it settles (Hopfield, 1982). Modern variants hold exponentially many patterns (Krotov & Hopfield, 2016; Ramsauer et al., 2021) and repair a corrupted cue as a side effect of converging. The obvious experiment—use one as an agent's persistent memory and compare—has not been run under matched conditions.

Why not. We think the answer is Proposition 1, and that it is the reason this comparison has been quietly circular wherever it has been attempted. At the operating point everyone actually uses, the two methods are the *same operator*. One step of the modern Hopfield update with the temperature turned up returns the top cosine neighbour exactly. Dense retrieval is not a rival sitting outside the energy formulation; it is a single corner of it. Any paper claiming that an energy-based memory beats similarity search must therefore say which coordinate it moved, and by how far—and if it ran at high β, it compared cosine against itself.

Stating the reduction first changes the question from *"is energy-based memory better?"*, which is not well posed, to *"which departures from the dense corner pay for themselves, and in which regime?"*, which is. It also makes every comparison in this paper exact by construction: the baseline is our own code with different arguments, so there is no implementation advantage to argue about and no encoder mismatch to suspect.

What we found. Mostly that the departures do not pay, and—more usefully—why. Of the mechanisms available beyond the corner, every one that changes only *how an answer is read off the landscape* fails on agent text: multi-step descent, residual peeling, repulsive parallel descent, vote-count readout, multi-scale and per-memory β. The two that help do not change the readout. One is a model of *what follows what*, fitted on the agent's own transitions. The other changes the *cue*: fold a few retrieved memories back into the query and retrieve again, which pays on exactly those questions whose second piece of evidence is reachable from the first rather than from the question, and pays nothing anywhere else. Neither is attractor dynamics, which is the finding.

We began with a sign that appeared to flip between modalities, and a mechanism that explained the flip. Energy descent computes a weighted average of stored patterns and moves the retrieval state onto their span. That should pay off when the query is a degraded copy of something already stored, because moving toward the memory manifold restores lost signal, and should cost accuracy when the query is not, because averaging smears the discriminative content of the cue. Question→evidence retrieval violates the premise and descent loses monotonically, as predicted. Image occlusion appeared to satisfy it, and descent won by more as damage increased.

That positive half did not survive the three tests we nominated against it. Masking spans of stored *text* satisfies the premise and yields no β that beats cosine in 45 swept cells. A cue that is a degraded copy of *two* stored images loses -0.245. And the image result itself, re-examined, is not completion at all: its metric is ten-way same-class accuracy, the cue's source image is never stored, and a soft k-NN class vote beats descent at every occlusion level, while the premise's own best case — occluded copies of images that *are* stored, scored by identity — is a null (§6.2). The premise now predicts every loss we measure and retains no confirming instance. §6 withdraws the sufficiency half and the image evidence rather than defending either.

Six further corpora then settled a question we could not answer from LoCoMo alone, and they settled it against us in the useful direction. *Completing a cue* from the memory bank does not work in any regime we have found. *Expanding a query* with retrieved memories does, by $+0.023$ to $+0.171$, on exactly those

multi-hop questions whose second document is reachable from the first rather than from the question, and by nothing at all on the two conversational-memory corpora this paper is about (§5.4).

Contributions.

C1 The reduction. Dense retrieval is exactly the (T=1, $\beta \to \infty$, b=0, R=I) corner of the modern Hopfield update (Proposition 1), verified at 1000/1000 queries on synthetic vectors and 838/838 on real sentence embeddings, both asserted in continuous integration. The equality is exact at k=1; at $k > 1$ the shortlist must be read off the *cue*, because ranks 2..k of the converged state order patterns by proximity to the winner instead (Remark 1) — which is not a caveat but the mechanism behind the shortlist regression we report. Every baseline in this paper is a configuration of the method under test. (X1)

C2 The β leverage band, and a hazard it implies. β has influence only within a narrow band, bounded below by collapse of the state to the corpus centroid on the first step and above by degeneration into cosine. Accuracy cannot see the upper edge—two retrievers can score identically while disagreeing about what they return—so we measure top-1 *decision* agreement with cosine, which rises to **99.4**% at β=512. A comparison run above the band therefore compares cosine against itself on nearly every query and will correctly report no difference for the wrong reason. The band is a property of the encoder, not a constant. (X3, X11, §3.4)

C3 A regime taxonomy, and the withdrawal of its one positive instance. Completion, composition and semantic match require mutually exclusive operating points, and the premise that separates them predicts a monotone −0.084 across an interpolation sweep on 1,977 LoCoMo questions. We then ran the three discriminating tests we had nominated against ourselves, and all three came back negative — including a matched-baseline audit of our own flagship image result, which a soft k-NN class vote beats by −0.036 [-0.043, -0.028]. The premise is reported as *necessary and not sufficient*, and the image result is withdrawn as evidence for an associative mechanism. (§6, §6.2, X113)

C4 Eight diagnosed negative results. Each is reported with the mechanism that produces it rather than as a bare null, including two that overturn claims we had previously made ourselves. (§5.5)

C5 One externally validated positive, on two public corpora, and a sharp limit on it. A nonparametric predictor over the agent's own observed transitions beats cosine on one-hop temporal association, in four measurements over three corpora. On LoCoMo it raises Recall@1 from 0.064 to 0.107 (+0.043, [+0.036, +0.050], $p < 10^{-4}$, +67% relative) over 5,233 probes in 272 sessions against a pooled 5,882-turn bank; the same driver, unmodified, gives +0.016 [+0.014, +0.018] on MSC over 27,815 probes against a 30,320-turn bank. At *two* hops LoCoMo returns −0.002 [-0.008, +0.004] and MSC +0.005 [+0.003, +0.006], a factor of 18 below the +0.185 two-hop figure of our own that we withdraw: association helps one step ahead and at best marginally thereafter. (§5.3, X26, X201)

C6 The measured price of seven protocol shortcuts. For each of seven errors we made, the magnitude of the false result it produced in our own data. Two were false *negatives*, which proved far harder to find than the false positives. The seventh is the one we would most want read: an *unmatched baseline* that every other safeguard in this paper was blind to, because split hygiene and pre-registration check how two arms are compared and not which two arms. (§7)

C7 One departure from the corner that *does* pay, and the question shape that predicts it. Folding K_0 retrieved memories back into the cue and retrieving again beats dense retrieval by +0.023 [+0.010, +0.037] on HotpotQA bridge questions and +0.171 [+0.118, +0.219] on 2Wiki inference questions, surviving a disjoint sample, a second encoder and two bank sizes, with random modes losing by +0.13 to +0.39. On comparison questions in the same corpora selection declines to expand in 10 of 10 folds. The predictor is hop type, not target count—our own pre-registered cardinality hypothesis comes out *reversed* on four corpora (Table 7) and we withdraw it. This is the one claim in the paper that contradicts the thesis we started with, it is query expansion rather than attractor dynamics, and it does not transfer across corpora. (§5.4, X202–X204)

What this paper is not. It is not a system paper, and it does not claim a better agent memory. The thesis we began with—that energy-based retrieval improves agent recall—is refuted here by our own measurements, at adequate power, with the mechanism identified. The one departure that does beat the corner (C7) is query expansion on a question shape we cannot yet detect without labels, it needs per-corpus tuning, and it is null on conversational memory, so it is a finding rather than a recommendation. We think the resulting object is more useful to the field than the system would have been: a reduction that makes the comparison exact, a taxonomy that says where descent loses and that kept none of the positive instance we had claimed for it, one sharply bounded positive that we did not expect and did not want, and a quantified account of how easy it is to publish the opposite conclusion.

2 Related Work

2.1 Agent memory systems: sophistication on the write path

MemGPT frames long-term memory as an operating-system paging problem, with the model managing transfers between a context window and external storage (Packer et al., 2023). MemoryBank adds a forgetting curve over stored summaries (Zhong et al., 2024). A-MEM constructs interlinked notes with agent-generated structure (Xu et al., 2025). HeLa-Mem introduces Hebbian co-activation and an episodic-to-semantic distillation stage (Zhu et al., 2026). These systems differ substantially in *what is written, summarised, linked and evicted*, and they are near-identical in how a memory is read: embed the query, take the top-k cosine neighbours.

HippoRAG is the most prominent attempt to move past that read operator, replacing flat similarity with personalised PageRank over a constructed knowledge graph (Gutiérrez et al., 2024). Its association structure is *spatial*—built from extracted entities and relations—where ours is *temporal*, learned from the order in which the agent's own episodes arrived, with no extraction or graph-construction stage. The two are complementary rather than competing, and we do not claim superiority over it; we have not run the comparison.

2.2 Associative and energy-based memory: sophistication on the read path

The classical Hopfield network stores patterns as minima of an energy function and recalls by descent (Hopfield, 1982), with capacity $\approx 0.14d$ before basins merge. Dense associative memory raises capacity with higher-order interactions (Krotov & Hopfield, 2016), and the continuous modern formulation attains exponential capacity with a one-step retrieval guarantee (Ramsauer et al., 2021). Universal Hopfield Networks place these variants in one framework of similarity, separation and projection (Millidge et al., 2022). Dynamic Manifold Hopfield Networks modulate attractor geometry by context (Li et al., 2025), and coupled-oscillator formulations separate stored patterns from a learned routing layer (Guo & Pasqualetti, 2026)—the latter directly motivating our decision to keep R separate from X in Equation (2).

This literature almost exclusively evaluates on synthetic patterns or image reconstruction. Our completion result (§5.1) is consistent with it, and §5.2 is the finding that does not transfer: the regime these papers study is the one in which the mechanism works, and agent memory retrieval is predominantly not that regime.

2.3 The equivalence, and why it has not been exploited

Ramsauer et al. (2021) prove that the modern Hopfield update is the attention mechanism of the transformer. This is usually read as a statement about transformers. Read in the other direction it says that a one-step, high-β Hopfield retrieval over a memory bank *is* an attention read over that bank—and, with normalised patterns and no strength term, is exactly cosine top-k. Proposition 1 is a corollary of their result rather than a new theorem, and we claim no priority for it. What we claim is the consequence: it is the reason the comparison between the two literatures has been circular wherever it has been attempted, and it should be stated as the first line of any such comparison rather than discovered in review.

2.4 Prediction in representation space

The joint-embedding predictive architecture proposes predicting the representation of unobserved content rather than the content itself (LeCun, 2022; Assran et al., 2023). Our module P (§3.5) is a JEPA-shaped predictor over an episode stream, and §5.3 is, to our knowledge, the first report that the *parametric* version of this fails in the low-data regime an individual agent's history actually provides (1.000 on fitted storylines, 0.000 on unseen), with a diagnosis—near-orthogonal displacements, no global transition direction—rather than a tuning complaint.

Predictive Associative Memory is the closest prior work: it retrieves by temporal co-occurrence using an external predictor alongside the memory (Dury, 2026). Our original claimed contribution against it was *integration*—folding the predictor into the energy functional instead of running it beside. §5.5 reports that this does not work, at any coupling strength, and withdraws the claim. What remains distinct is the reduction frame, the regime taxonomy, and the nonparametric form of the predictor.

2.5 Mixtures, sampling, and statistical practice

The rank argument for mixtures of softmaxes (Yang et al., 2018a) establishes that our integrated multi-β variant is not reducible to any single β; we confirm the expressiveness empirically and find it performance-neutral (Table 9). The stochastic relaxation we specify and then decline to build on rests on Boltzmann machines (Ackley et al., 1985) and Langevin sampling (Welling & Teh, 2011). Residual peeling is matching pursuit (Mallat & Zhang, 1993) over the energy landscape, and repulsive parallel descent is Stein-variational in flavour (Liu & Wang, 2016). Our interval procedure follows the clustered bootstrap (Cameron et al., 2008); the headroom that makes any of these measurable at all on text is a consequence of embedding anisotropy (Ethayarajh, 2019; Gao et al., 2019), without which, as §4 notes, cosine saturates and nothing can be compared.

3 The Reduction

3.1 Setup and notation

Memories $m_1, \dots, m_N$ are embedded by a frozen encoder E into unit vectors $z_i = E(m_i) \in \mathcal{S}^{d-1}$ and stacked as $X = [z_1 \cdots z_N] \in \mathbb{R}^{d \times N}$. A cue x is embedded to $q = E(x)$, and the retrieval state is initialised at $\boldsymbol{\xi}^{(0)} = q$. Write $\Pi[\cdot]$ for projection back to the sphere, $b \in \mathbb{R}^N$ for a per-memory strength bias, $\beta > 0$ for an inverse temperature, and $\mathrm{lse}(\cdot)$ for the log-sum-exp. The modern Hopfield energy (Ramsauer et al., 2021) is

$$E_M(\boldsymbol{\xi}) \;=\; -\tfrac{1}{\beta}\,\mathrm{lse}\big(\beta(X^\top \boldsymbol{\xi} + b)\big) \;+\; \tfrac{1}{2}\|\boldsymbol{\xi}\|^2, \tag{1}$$

minimised by the concave–convex (CCCP) update, which we run for T steps:

$$\boldsymbol{\xi}^{(t+1)} \;=\; \Pi\Big[(1-\eta)\, X\ \mathrm{softmax}\big(\beta(X^\top \boldsymbol{\xi}^{(t)} + b)\big) \;+\; \eta\, R\big(\boldsymbol{\xi}^{(t)}\big)\Big]. \tag{2}$$

Equation (2) decreases (1) monotonically and converges for $\eta = 0$. The operator R is a *routing* map, discussed in §3.3.

3.2 Dense retrieval is a corner of this family

Proposition 1 (The reduction)**.** *Let the patterns z_i and the cue q be L_2-normalised, and set $T = 1$, $b = 0$, $\eta = 0$. Then as $\beta \to \infty$ the state after one update converges to the stored pattern of maximal cosine similarity to the cue,*

$$\lim_{\beta \to \infty} \Pi\big[X\ \mathrm{softmax}(\beta X^\top q)\big] \;=\; z_{i^\star}, \qquad i^\star = \arg\max_i \langle z_i, q\rangle,$$

whenever the maximiser is unique. The reduction is therefore exact *at $k = 1$. For $k > 1$ the dense top-k neighbourhood is recovered by reading the ranking off the* cue*, $\langle z_i, q\rangle$ — the standard definition, and the one our dense baseline uses — and* not *off the converged state: ranks 2..k of $\langle z_i, \boldsymbol{\xi}^{(1)}\rangle$ order the patterns by*

proximity to $z_{i^\star}$, not to q, because the state has forgotten the cue by construction. Remark 1 makes this precise, and §5.1 shows it is the mechanism behind the shortlist regression rather than a technicality.

Proof sketch. With $b = 0$ the logits are $s_i = \langle z_i, q \rangle$. Let $\delta = s_{i^\star} - \max_{i \neq i^\star} s_i > 0$. Then $\mathrm{softmax}(\beta s)_{i^\star} = \big(1 + \sum_{i \neq i^\star} e^{-\beta(s_{i^\star} - s_i)}\big)^{-1} \geq (1 + (N-1)e^{-\beta\delta})^{-1} \to 1$, and every other coordinate is $O(e^{-\beta\delta})$. Hence $X\,\mathrm{softmax}(\beta X^\top q) \to z_{i^\star}$, and normalisation fixes the point since $\|z_{i^\star}\| = 1$. The full argument — with the explicit $O\big((N-1)e^{-\beta\delta}\big)$ rate, the tied-maximiser case, and the analysis of what happens for $k > 1$ — is in Appendix A. □

We treat Proposition 1 as an executable claim rather than a remark. The test suite instantiates the configuration above and asserts *top-1* agreement with an independent exact-cosine implementation on every query — the quantity the proposition states, no more; it passes 1000/1000 on synthetic vectors (d=64) and 838/838 on real sentence embeddings (d=768), to floating-point precision, and it gates continuous integration. If it ever fails, no downstream number in this paper is trustworthy (X1).

Two consequences organise everything that follows.

First, the baseline cannot be disadvantaged. "Dense top-k" in every table below is Equation (2) with the arguments of Proposition 1, run through the same code path, the same encoder and the same candidate set as the method under test. The most common rejection reason for papers of this shape—that the authors tuned their own method and not the baseline—is unavailable here by construction.

Second, the research question becomes finite. The reduction identifies a specific point in a parameter space; the scientific content is which coordinates are worth moving.

3.3 Three axes away from the corner

Exactly three coordinates of Equation (2) are free, and each encodes a distinct hypothesis about what agent memory needs.

G1 — depth, $T > 1$. Iterating repairs a degraded cue. Cosine cannot imitate this: it has no mechanism for revisiting a cue.

G2 — basin depth, $b \neq 0$. A memory used constantly earns a deeper basin and therefore a wider one, so a vaguer cue still finds it. This folds usage, recency and surprise into a single term of the energy instead of three heuristics blended by hand outside it. Cosine can partially imitate it with an additive score bonus.

G3 — routing, $R \neq I$. Memories that co-occur in time pull on one another, so falling into one drags the state toward its companions. This is the only axis that can express a relation cosine structurally cannot represent, and it is the axis on which our one positive result lives. Critically, R must be fitted from temporal co-occurrence and never from embedding similarity: fitting it on similarity collapses G3 back into the baseline and silently invalidates the claim.

3.4 The β leverage band

Proposition 1 has a corollary that is easy to state and, we will argue, widely consequential.

Corollary 1 (High-β degeneracy)**.** *For any T, as $\beta \to \infty$ with $b = 0, \eta = 0$ the trajectory of Equation (2) is constant after the first step, because stored patterns are exact fixed points. Multi-step descent at high β is therefore* identical *to one-step descent, which by Proposition 1 is cosine retrieval.*

Empirically the degeneracy arrives early, and accuracy is the wrong way to see it. Two retrievers can score identically while disagreeing about which memory they return, so we measure *top-1 decision agreement* with cosine — the fraction of queries on which descent returns the exact memory cosine would have (Figure 1, `sweep_beta_transition.py`).

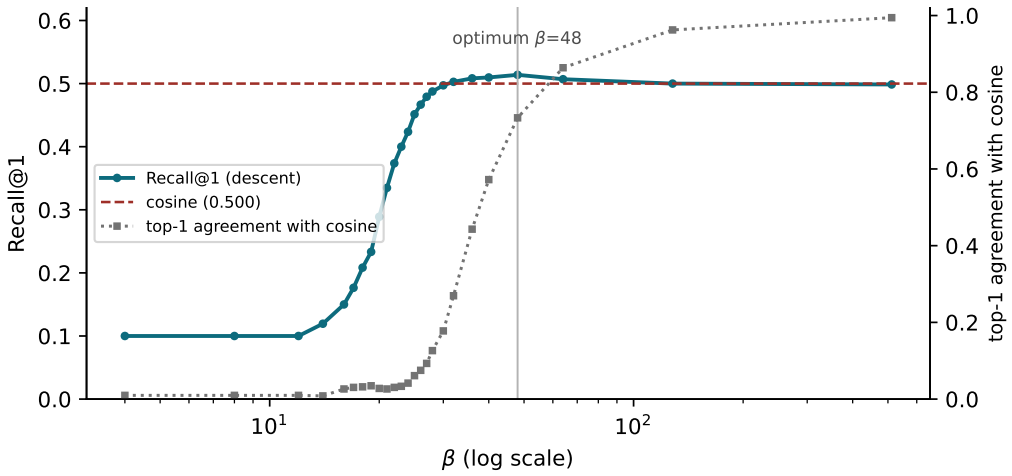


Figure 1: The β leverage band, on the held-out image task. **Left axis:** same-class top-1 accuracy against the cosine baseline (dashed); the metric is ten-way classification, which is why the collapse floor is $0.100 = 1/10$ (§6.2). **Right axis:** the fraction of queries on which descent returns the *same memory* as cosine. Below the band the state collapses to the corpus centroid; above it descent makes cosine's decision on 99.4% of queries. The interval in between is where β has any influence at all — which is not the same as where it is useful: §6.2 shows the interior optimum is beaten by a classifier.

The curve has two failures and one working region. Below $\beta \approx 24$ the softmax is flat enough that the first step averages a large fraction of the bank and the state lands at its centroid: same-class accuracy is 0.100 at β=12, which is exactly chance for a ten-class bank, and decision agreement with cosine is 1%. Above the band, descent converges on cosine's own answer — agreement rises through 86% at β=64 and 96% at β=128 to **99.4**% at β=512, where accuracy is 0.4986 against cosine's 0.5000. Between them is a genuine interior optimum: β=48 reaches 0.514, +0.014 over cosine, while still disagreeing with it on 27% of queries. The band is narrow — roughly $[32, 64]$ on this encoder, with leverage over perhaps a factor of two in β. We now read that interior optimum as the width at which the blend acts as a class vote rather than as an attractor: at β=40 the first step has an effective support of 12.7 patterns out of 400 (X113). The band is where β matters; it is not where the method wins.

Three implications. (i) The apparent *plateau* in accuracy above $\beta \approx 64$, which a coarse sweep reads as a robust operating region, is the method dissolving into the baseline rather than working; the agreement curve distinguishes the two and the accuracy curve cannot. (ii) The band is a property of the encoder and must be re-measured per encoder: the interval above is for ResNet-18 features (d=512), while on `all-mpnet-base-v2` (d=768) the lower edge sits higher — at β=16 the effective number of patterns in the blend is 377 (X11), against a threshold of ≈ 16 on isotropic synthetic vectors (X3). The threshold survives as a mechanism and dies as a number. (iii) Any experiment comparing "Hopfield retrieval" against "cosine retrieval" at high β is comparing cosine against cosine on nearly every query, and will correctly report no difference for the wrong reason. We flag this as a live hazard for this literature; we committed it ourselves, and §7 reports what it cost us.

3.5 Beyond the landscape: memory as a model of what follows what

Every mechanism in §3.3 except G3 changes how an answer is *read off* a fixed landscape. None of them changes the landscape. The results in §5.5 make that distinction empirically load-bearing: readout mechanisms uniformly fail on agent text.

G3 is different in kind, and the formulation we adopt for it is the joint-embedding predictive one (LeCun, 2022; Assran et al., 2023): predict the *representation* of the next episode, not its tokens. An agent's episode

stream is a trajectory, and temporal adjacency is free supervision—no labelling is required to know which episode followed which.

Let $\mathcal{T}$ be the set of observed consecutive index pairs, Z_c the matrix of cue representations over $\mathcal{T}$, and $\Delta_i = z_{\text{next}(i)} - z_{\text{cue}(i)}$ the displacement each transition took. We consider two instantiations.

Parametric. A low-rank predictor trained with an InfoNCE objective against a frozen target encoder. This is the natural first choice and it fails completely (§5.3): it memorises the storylines it was fitted on.

Nonparametric. Carry the transitions rather than compressing them, and transfer displacements locally:

$$P(q) \;=\; \Pi\Big[q \;+\; \alpha \textstyle\sum_{i \in \mathcal{N}_k(q)} w_i \, \Delta_i\Big], \qquad w = \text{softmax}\left(\tau \, Z_c^\top q\right)\big|_{\mathcal{N}_k(q)}, \tag{3}$$

where $\mathcal{N}_k(q)$ are the k nearest remembered cues and α is a step size. This is still a world model—it predicts what comes next in embedding space—and it composes: iterating it rolls the model forward. It is the one mechanism in this paper that beats cosine on held-out data.

The step size α is not cosmetic, and §7 explains why we are emphatic about it. A single remembered displacement has norm 1.157 against a unit-norm cue, so at α=1 the prediction is dominated by the displacement and lands essentially on $z_{\text{next}(i)}$—a *stored training episode*, 100% of the time on our corpus against a bank prior of 14%. The kernel in (3) implicitly applies $\alpha \approx 0.34$ through cancellation among near-orthogonal displacements, and that shrinkage, not the weighting, is what keeps the cue in control of where the prediction lands (X23B). Any comparison between predictors that does not match effective step size measures step size.

4 Experimental Protocol

Protocol decisions are listed here in full because several of them changed a conclusion's sign, and §7 reports by how much.

Matched encoder. Every method under comparison—ours, every baseline, every ablation—is handed identical vectors from the same frozen model. Text uses `all-mpnet-base-v2` (Reimers & Gurevych, 2019), d=768; images use ImageNet-pretrained ResNet-18 features (He et al., 2016), d=512. Encoders are never fine-tuned, and no result in this paper can be attributed to representation quality.

Corpora.

- **Images (completion regime).** 400 Imagenette images (40 per class, ten classes) from the `train` split as the memory bank; queries are occluded images from the `val` split at 30/50/70% occlusion. Evaluation uses 5,847 query instances drawn from 1,949 distinct images, the full validation split. **Two properties of this design are load-bearing and were missing from earlier versions of this paper.** The splits are disjoint, so a cue's own source image is *not* in the bank; and the metric is **same-class** top-1 accuracy — a retrieval is correct when the returned memory shares the query's class — so the task is ten-way classification through a retrieval interface, with chance at 0.100. §6.2 reports what follows from both facts, and X113B supplies the same-image variant in which the cue is an occluded copy of a pattern that *is* stored.
- **LoCoMo (semantic-match regime).** 1,977 real questions over 5,882 conversational turns (Maharana et al., 2024). Metric: Coverage@5 over gold evidence turns.
- **LoCoMo (composition regime).** The same 5,882 turns, re-used as temporal-association probes: cue a turn, retrieve the turn that followed it in the same session. 5,233 probes at one hop and 5,013 at two, over 272 sessions. The candidate set is the full pooled bank of all ten conversations, with dialogue identifiers namespaced by conversation (Appendix E). This is the external-validity evidence for §5.3, and it is a corpus we did not build.

- **Episodic corpus (composition regime).** A purpose-built store of 1,319 episodes across 122 storylines with 146 multi-target questions, plus 900 compositional filler episodes and planted near-miss distractors. Association probes are generated from it as consecutive (or h-distant) within-storyline pairs: 257 probes at one hop and 135 at two. This corpus is ours, it is generated rather than collected, and on its own it would not support a claim; every composition result measured on it is reported beside the LoCoMo measurement of the same quantity.
- **MSC (composition regime).** 30,320 pooled turns over 2,505 sessions and 501 dialogues (Xu et al., 2022), run through the *unmodified* LoCoMo association driver so that the only thing that changes is the corpus. This is the second public corpus for §5.3 and the largest bank in the paper (X201).
- **Five retrieval corpora (composition regime, question→evidence).** HotpotQA (Yang et al., 2018b) and 2WikiMultihopQA (Ho et al., 2020), both of which label the *shape* of each question; QASPER (Dasigi et al., 2021), whose banks are single papers of ≈47 paragraphs; and SciFact and NFCorpus from BEIR (Thakur et al., 2021). 2,000 questions each for the two multi-hop corpora over banks of ≈995 and ≈800 paragraphs, 812 for QASPER, 300 and 323 for the two BEIR tasks. Metric: gold@10, the fraction of a question's gold evidence recovered in the top ten. These corpora exist to answer one pre-registered question — whether LoCoMo's null for query expansion is a property of the method or of LoCoMo — and §5.4 reports that it is the latter (X202–X204).
- **LongMemEval-S (semantic-match regime).** 479 chat-assistant memory questions, each with its own ≈493-turn haystack (Wu et al., 2025), with the question-type strata the benchmark ships. The second conversational-memory corpus, and the one that decides whether LoCoMo's pattern is LoCoMo's alone (X200).
- **Two further modalities (composition regime).** 2,000 2×2 image mosaics assembled from $m \in \{1, \ldots, 4\}$ bank images, where m is set by construction rather than inferred (X211); and Flickr8k under CLIP ViT-B/32 (Radford et al., 2021), giving a text→image arm at m=1 and an image→text arm at m=5 (X212). Both exist to test one hypothesis of ours about target cardinality, and Table 7 reports what they did to it.

A second encoder, and a note on what that buys. The five retrieval corpora above were re-run under `bge-base-en-v1.5` (Xiao et al., 2024) as a robustness check. We report it as a check and not as a second experiment: the matched-encoder rule holds *within* every comparison, so a second encoder tells us whether a conclusion is an artefact of `mpnet`, and §5.4 reports one conclusion that is.

Baselines. Dense top-k (Proposition 1), BM25 (Robertson & Zaragoza, 2009), hybrid reciprocal-rank fusion (Cormack et al., 2009), and—in the composition regime—a parametric JEPA predictor. All lexical and hybrid baselines receive the same tuning budget as our methods, stated per experiment in Appendix B.

Splits and selection. Hyperparameters are never selected on the data a number is reported on. In the composition regime the split unit is the *storyline* on our corpus and the *session* on LoCoMo, never the question: a model fitted on transitions from one storyline or session is never evaluated on it. Because the corpus holds only 122 storylines, a single 50/25/25 split leaves 61 one-hop and 30 two-hop test probes, intervals on which span most of the plausible range before the experiment begins. We therefore use thread-level 4-fold rotation: every storyline is in the test set exactly once and is never in the train or validation fold that tests it. Both the world model's own (k, τ) and the readout's parameters are selected per fold on that fold's validation storylines.

Intervals. Headline numbers are *paired* differences against the baseline on the same queries, with 95% bootstrap intervals (20,000 resamples) over the *cluster*—questions, storylines, or source images as appropriate—never over individual measurements. Comparing two overlapping marginal intervals is the wrong test when both methods see the same queries; resampling rows rather than clusters is pseudo-replication and we quantify what it cost us in §7. Image-modality intervals resample *images*, not query instances, since instances from one image are correlated across occlusion levels and seeds.

Table 1: The image study with its matched baseline. The metric is **same-class top-1 accuracy** over ten classes, not identification, and the cue's source image is *not* in the bank — see §6.2. β, k and τ all selected on the same disjoint tune half; one pre-committed paired test on the held-out half. n=5,847 query instances from 1,949 distinct images; bootstrap resamples images. *CI excludes zero.

Occlusion	**Cosine**	**Descent** (β=40)	**k-NN vote** (k=20)	**Descent − vote (95% CI)**
30%	0.730	0.733	0.761	−0.028 [-0.040, -0.017]*
50%	0.542	0.559	0.600	−0.041 [-0.054, -0.027]*
70%	0.386	0.407	0.445	−0.038 [-0.052, -0.024]*
pooled	**0.553**	**0.566**	**0.602**	**−0.036** [-0.043, -0.028]*

Table 2: Completion as the premise defines it (X113B): cues are occluded copies of images that *are* in the bank, scored by **same-image** recall@1. Identity control at 0% occlusion: cosine recall@1 = 1.0000. β selected on a disjoint half of the bank images and used once; selection chose β=80, the top of the grid. Same-class accuracy on the identical cues is shown for comparison with Table 1.

	Same-image recall@1			**Same-class accuracy**		
Occlusion	Cosine	Descent	Δ (95% CI)	Cosine	Descent	Δ
30%	0.855	0.855	+0.000 [+0.000, +0.000]	0.915	0.915	+0.000
50%	0.635	0.630	−0.005 [-0.015, +0.000]	0.755	0.760	+0.005
70%	0.380	0.375	−0.005 [-0.015, +0.000]	0.545	0.535	−0.010
pooled	**0.623**	**0.620**	**−0.003** [-0.008, +0.000]	0.738	0.737	−0.002

Dimensionless hyperparameters. Every scale that crosses dimensionalities is normalised before use: a perturbation of nominal magnitude σ is applied as $\sigma/\sqrt{d}$, so that its displacement norm is dimension-independent. At d=768 the unnormalised form is wrong by a factor of 28.

Symmetric candidate sets. Where a cue is itself a stored memory—as in every association probe—it is excluded from the candidate set for *every* method identically, before ranking. Asymmetry here produces exact zeros that read as findings.

Latency. No accuracy number appears without its cost. Iterative descent at T=8 costs 21× cosine in measured wall-clock (0.0502 ms vs. 0.0024 ms per query), and multi-β ensembling a further 1.6–2.9×. We report both wall-clock and matmul counts, because they diverge: the integrated mixture issues the same 16 matmuls as single-β descent and costs 1.6× the wall-clock, since mixing m softmaxes per step is arithmetic outside the matmuls. Figures in Table 9 and Appendix B.

Record. Every experiment is logged in an append-only ledger with an identifier, the question it asked, the result, and what it changed. Entries are never edited after their result is in; an overturned entry becomes a new entry pointing back at it. Identifiers of the form Xn in this paper index that ledger, including the entries that overturn our own earlier claims.

5 Results

All numbers below are paired against the dense baseline of Proposition 1 on identical queries, with cluster-bootstrap intervals. Ledger identifiers index `docs/10_EXPERIMENT_LEDGER.md`.

5.1 Completion: descent beats cosine, and loses to a plain classifier

Table 1 was for most of this project the paper's one clean positive for the energy mechanism itself, and the first three columns are unchanged from that version: descent beats cosine by +0.014 pooled, growing with

Table 3: LoCoMo, 1,977 questions over 5,882 turns. `anchor` interpolates between pure cosine (1.0) and pure descent (0.0); the baseline is a point in our own parameter space, so the sweep measures exactly what descent adds. The reduction check passes 196/196 at `anchor`=1.0.

Condition	Coverage@5	95% CI
`anchor`=1.00 ***(is cosine)***	**0.336**	[0.316, 0.355]
`anchor`=0.90, pp_0=32	0.331	[0.311, 0.351]
`anchor`=0.75, pp_0=4	0.321	[0.301, 0.340]
`anchor`=0.50, pp_0=4	0.304	[0.285, 0.324]
`anchor`=0.25, pp_0=4	0.273	[0.255, 0.292]
`anchor`=0.00, pp_0=4	0.252	[0.234, 0.271]
energy, fixed β=32	0.236	[0.218, 0.254]

corruption severity, on a pre-committed held-out split. The fourth column is why we no longer report it that way. A soft k-NN class vote — selected on the same tune half, from a grid declared in advance — scores 0.602 where descent scores 0.566, and beats descent at every occlusion level. §6.2 gives the mechanism: on a *class* metric, descent at β=40 is an implicit class vote over ≈ 13 neighbours, and an explicit vote over k=20 does the same job better. Table 2 then asks the question the metric name implied all along — can descent retrieve the *particular* stored image a cue was degraded from — and the answer is -0.003 [-0.008, +0.000].

The latency comparison is unchanged and now lands differently: descent costs 21× cosine's wall-clock latency (0.0502 ms vs. 0.0024 ms per query) to be beaten by a classifier that costs one extra argpartition.

Two further qualifications, which predate X113 and survive it. First, the apparent advantage is *capacity-dependent*: growing the bank from 100 to 4,000 patterns at fixed β=32, cosine overtakes descent past $N \approx 800$. Second, descent's rank-1 gain over cosine comes with a *shortlist regression*: recall@5 is 0.605 against cosine's 0.759 and MRR 0.589 against 0.643, all class-level (Table 9). Converging to a single attractor means ranks 2–5 measure proximity to that attractor rather than to the query, which is Remark 1 and is now confirmed in three independent settings.

Two earlier verdicts of ours, withdrawn. A previous version of this study used 80 distinct images with unpaired intervals roughly 5× the effect size and β tuned on the test images; those conclusions were underpowered and Table 1 supersedes them. The version that replaced it — the one with the honest split, which we described as a clean positive for the energy mechanism — is withdrawn in turn by its own fourth column. The number stands; the interpretation does not.

5.2 Semantic match: every increment of descent costs accuracy

Table 3 is monotone: the optimum sits at the boundary where the method *is* the baseline. The table shows the best pp_0 at each anchor; the worst cell of the full sweep is `anchor`=0.00, pp_0=32 at 0.201 [0.184, 0.218], which is the broad-start annealing failure of §3.4 compounding with the absent query anchor. There is no interior operating point at which descent helps on question→evidence retrieval, which is the dominant query type in deployed agent memory.

A data bug that invalidated an earlier result. LoCoMo dialogue identifiers are unique only *within* a conversation: 5,882 turns carry 1,033 distinct identifiers, and 132 of conversation 0's 134 gold identifiers recur in other conversations. Our first capacity sweep pooled banks without namespacing, so distractors satisfied gold identifiers and scored as hits. Its headline—"adaptive β improves as the bank grows"—was an artefact; with namespaced identifiers adaptive β *degrades* (0.250 → 0.231), merely less steeply than fixed β (0.244 → 0.216). We report this because the failure mode is specific to this widely used benchmark and will recur.

Table 4: One-hop temporal association. Four measurements, three corpora. "Clusters" is the bootstrap resampling unit—storylines for the episodic corpus, sessions for LoCoMo and MSC. Implementations (ii) and (iii) were written without reference to (i); (iv) is (iii)'s driver run unchanged on a different corpus. Absolute scores fall as the bank grows (1,319 → 5,882 → 30,320 candidates), which is why MSC has the smallest relative effect and the narrowest interval: it is the hardest row and the best powered.

	Corpus	Probes	Clust.	Cosine	WM	Paired Δ R@1
(i)	episodic (ours), 50/50	123	61	0.106	0.130	+0.024 [-0.032, +0.083], p=0.24
(ii)	episodic (ours), 4-fold	257	122	0.105	0.140	+0.035 [+0.012, +0.061], p=0.002
(iii)	**LoCoMo**, 4-fold	**5,233**	**272**	**0.064**	**0.107**	**+0.043** [+0.036, +0.050], $\mathbf{p < 10^{-4}}$
(iv)	**MSC**, 4-fold	**27,815**	**2,505**	**0.049**	**0.065**	**+0.016** [+0.014, +0.018], $\mathbf{p < 10^{-4}}$

5.3 Composition: the one mechanism that works

The gate. Before building a predictor we checked whether cosine already solves "what happened next". It does not: Recall@1 is 0.105 with the gold episode at median rank 24 of 1,319. The failures are instructive—they are lexical collisions across unrelated storylines, where a cue about a cottage drive returns "forty minutes each way" from an apartment-hunting thread. Oracle-restricting candidates to the correct storyline still reaches only 0.488, so the difficulty is not merely retrieval breadth. Temporal succession is genuinely not a similarity relation (X16).

The parametric predictor fails completely. A low-rank JEPA predictor fitted on some storylines and evaluated on others reaches Recall@1 of 1.000 on the storylines it was fitted on and 0.000 on unseen ones, against the cosine baseline's 0.131—worse than doing nothing. No capacity escapes it: across ranks 2 to 32 the best unseen Recall@1 at any setting is 0.016, and the two largest capacities score exactly zero. The diagnostic explains why: displacement vectors between consecutive episodes are near-orthogonal (mean pairwise cosine +0.004) and the top ten principal components carry only 31% of their variance. There is no global "and then" direction to fit, and 134 transitions in 768 dimensions cannot teach one (X17, re-run as X17-R). We report this as a result about inductive bias, not as a tuning failure.

The nonparametric predictor works, on our corpus and on a public one. Equation (3), with (k, τ) selected on held-out storylines or sessions only, beats cosine on one-hop association in four independent measurements over three corpora (Table 4). The last two are the ones that matter: they run the same probe design on LoCoMo and on MSC, benchmarks built by other groups. On LoCoMo the run is against the *full pooled bank* of 5,882 turns from all ten conversations, so a prediction must beat distractors from nine unrelated conversations as well as its own. Dialogue identifiers are namespaced by conversation—without that, 5,882 turns carry only 1,033 distinct identifiers and distractors satisfy gold identifiers, the bug described in §5.2.

On LoCoMo the effect is +67% relative at Recall@1 and +0.077 [+0.067, +0.087] at Recall@5 (0.174 → 0.251).

It replicates on a second public corpus, five times the size. Row (iv) of Table 4 runs the *unmodified* LoCoMo driver on MSC (Xu et al., 2022): 27,815 probes over 2,505 sessions against a pooled bank of 30,320 turns, more than five times LoCoMo's. Recall@1 rises from 0.049 to 0.065, $\Delta = +0.016$ [+0.014, +0.018], $p < 10^{-4}$, +33% relative; Recall@5 gains +0.028 [+0.026, +0.031] (X201). The absolute numbers are lower and the relative effect is smaller than LoCoMo's, which is what a bank five times larger should do to a Recall@1. The direction, the significance and the mechanism are the same, on a corpus we did not build, with code we did not change. This is the paper's load-bearing positive claim and it is now the only one measured on two independent public corpora.

The mixture readout is again below plain averaging on MSC (+0.006 against averaging's +0.016), reproducing the LoCoMo ordering of §5.6 — a fourth corpus on which preserving the modes is worse than averaging them.

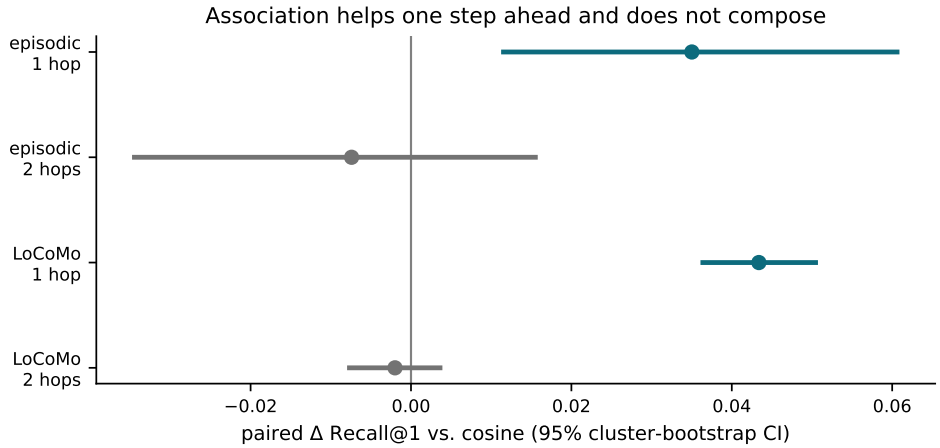


Figure 2: Paired Δ Recall@1 against cosine, one and two hops, for the two corpora the figure was built from (the episodic corpus and LoCoMo; MSC is in Tables 4 and 5 and is not plotted here). Intervals are 95% cluster bootstrap over storylines and sessions respectively. The one-hop effect is positive on both and larger on the public, harder corpus; the two-hop effect is null on both. Generated by `scripts/make_tables.py`.

Table 5: Two-hop composition, every measurement we have. The original script was lost before it was committed, so (a) cannot be re-run; (b), (c) and (d) are independent reimplementations under storyline- and session-disjoint rotation. Row (c) is p=0.77. Row (d) excludes zero at n=25,310 but at 1/18 the magnitude of the withdrawn +0.185; the original claim is withdrawn and the surviving effect is too small to build on.

	Corpus / protocol	Probes	Metric	Result
(a)	episodic, 50/50 *(lost)*	65	R@5	0.446 vs. 0.262; Δ+0.185
(b)	episodic, 4-fold	135	R@5	0.230 vs. 0.207; spans zero
(c)	LoCoMo, 4-fold	5,013	R@1	0.041 vs. 0.043; $\Delta - 0.002$ [-0.008, +0.004]
(d)	MSC, 4-fold	25,310	R@1	0.017 vs. 0.012; $\Delta + 0.005$ [+0.003, +0.006]

Multi-hop composition does not hold. We withdraw our own earlier result. An earlier run of ours indicated that iterating the predictor recovers two-hop chains it was never trained on: Recall@5 of 0.446 against cosine's 0.262, $\Delta = +0.185$ $[+0.076, +0.297]$, $p < 0.001$. That was the strongest result we had recorded. It does not survive.

Three subsequent measurements (Table 5) return nulls, the last of them on public data at n=5,013 with 272 clusters—roughly 37× the probe count of the original. We therefore withdraw the two-hop claim as an artifact rather than suspending it. The +0.185 figure should not be cited, including from our own ledger, where it now carries a pointer to this result.

A residual signal does exist at two hops on LoCoMo at Recall@5 (+0.012 $[+0.004, +0.019]$), an order of magnitude below the one-hop effect and not visible at Recall@1.

MSC makes the two-hop picture less clean, and we report it that way. Row (d) of Table 5 is *positive*: Δ Recall@1 = +0.005 $[+0.003, +0.006]$ at n=25,310, with Recall@5 at +0.010 $[+0.008, +0.012]$. The interval excludes zero, so this is not a null. It is also 3.4× smaller than the same corpus's one-hop effect and 9× smaller than LoCoMo's, and it sits at an absolute Recall@1 of 0.017. Our reading is that a second hop carries a real but very small amount of signal that 25,310 probes can resolve and 5,013 cannot — which is a statement about power, not a rescue of the withdrawn claim. The withdrawn figure was +0.185 at Recall@5; what survives is +0.010, a factor of 18 below it.

We therefore state the summary with its qualification attached: association helps one step ahead, and at two steps it either vanishes or shrinks to a margin too small to build on, because the error introduced by

Table 6: Two-stage expansion across six corpora, stratified by question shape. Metric is gold@10. The **Null** column is the number of folds in which selection chose K_0=0, i.e. declined to expand: where it reads 10/10 the +0.000 is the method refusing to act, not a measurement that came out at zero, and no interval can be placed on it. * marks an interval excluding zero. The pattern is not about corpus and not about target count — it is about whether the next piece of evidence is reachable from the retrieved evidence ("chain") or only from the question. RRF's column is given for scale and is discussed below.

Corpus	Question shape	n	Null	Cosine	Exp.	Expansion − cosine	RRF
HotpotQA	bridge (chain)	1,591	0/10	0.889	0.912	+0.023 [+0.010, +0.037]*	+0.035
	comparison	409	10/10	0.976	0.976	+0.000 [+0.000, +0.000]	+0.017
2Wiki	inference (chain)	246	0/10	0.724	0.894	+0.171 [+0.118, +0.219]*	+0.077
	compositional (chain)	834	0/10	0.704	0.770	+0.067 [+0.042, +0.091]*	+0.005
	comparison	463	10/10	0.916	0.916	+0.000 [+0.000, +0.000]	+0.080
	bridge-comparison (m=4)	457	0/10	0.560	0.559	−0.001 [-0.023, +0.021]	+0.043
QASPER	single-evidence	584	0/10	0.627	0.663	+0.036 [+0.005, +0.068]*	+0.103
	multi-evidence	228	8/10	0.505	0.478	−0.027 [-0.049, -0.010]*	+0.004
SciFact	all	300	0/10	0.790	0.803	+0.013 [-0.014, +0.040]	+0.042
NFCorpus	multi-relevant (m≈41)	300	0/10	0.149	0.166	+0.017 [+0.008, +0.027]*	+0.009
LongMemEval	all	479	1/10	0.639	0.632	−0.007 [-0.032, +0.018]	+0.156
	temporal-reasoning	132	0/10	0.483	0.464	−0.019 [-0.086, +0.048]	+0.255

predicting from a prediction grows faster than the compositional signal. Any reader who wants the two-hop effect called positive on MSC is entitled to; nothing in this paper rests on which way that is called.

Why we report this at length. The withdrawn result was, for several weeks, the headline of this project and the organising claim of a design document built on top of it. It was overturned by reimplementation, not by review—twice—and the version that finally settled it is the public-benchmark run that a reviewer would have demanded anyway. We regard the sequence as the strongest available evidence for §7's argument that independent reimplementation is a distinct and stronger property than peer inspection.

5.4 Six more corpora, and the one departure that does pay

Everything to this point says the departures from the dense corner do not pay. One pre-registered objection to that was still open, and it is the obvious one: every text number above comes from LoCoMo. On LoCoMo, two-stage query expansion — retrieve K_0 modes, fold them into the cue at temperature β, retrieve again — is selected straight to its null, K_0=0. We could not tell whether that is a fact about the mechanism or a fact about the corpus. The mean mutual cosine between a LoCoMo question's gold evidence turns is 0.527 — the pieces of evidence already look like each other, so a mode retrieved from one is close to what the cue would have found anyway. On a corpus built so that the second piece of evidence is reachable only *from the first*, expansion should have something to do.

Protocol. `PREREG_HI.md` was committed *alone*, before any of these corpora was downloaded, which is the fix for the earlier pre-registration whose file landed in the same commit as its results (§7). Six corpora go through the unchanged harness: leave-one-group-out selection over ten groups on the training side only, per-fold picks recorded in the JSON, cluster bootstrap over banks. One deviation is logged in the pre-registration: the robustness checks below were chosen *after* seeing the first result, so they are exploratory and labelled as such.

The result. Table 6. On chain-shaped questions, expansion wins, and this is the first time in this project that a departure from the corner has: +0.023 [+0.010, +0.037] on HotpotQA bridge questions, +0.171 [+0.118, +0.219] on 2Wiki inference and +0.067 [+0.042, +0.091] on 2Wiki compositional. On comparison questions — where both documents are reachable from the question directly — selection chooses K_0=0 in 10 of 10 folds on both corpora, and the method correctly does nothing. On QASPER, whose banks are single papers of ≈47 paragraphs, expansion actively hurts the multi-evidence stratum, −0.027 [-0.049, -0.010]: with

Table 7: Every pre-registered test of our own prediction that more targets should make expansion help more. Four reversals, one confirmation, two nulls. Two rows rest on a 23-query stratum (SciFact multi-relevant, NFCorpus single-relevant) and should be read as weak in both directions; NFCorpus, the one confirmation, is one of them. The mosaic row is exactly zero because the method selected K_0=0 in 10/10 folds at every m, so the interaction is zero by construction.

Corpus	Contrast	n	Interaction (95% CI)	Verdict
2Wiki	m=4 − m=2	457 / 1,543	−0.023 [-0.044, -0.004]*	reversed
QASPER	multi − single	228 / 584	−0.036 [-0.070, -0.001]*	reversed
SciFact	multi − single	23 / 277	−0.025 [-0.050, -0.004]*	reversed
NFCorpus	multi − single	300 / 23	+0.011 [+0.002, +0.022]*	as predicted
LongMemEval	multi-sess. − single	125 / 150	+0.002 [-0.040, +0.044]	null
Flickr8k	m=5 − m=1	1,000 / 5,000	−0.008 [-0.013, -0.004]*	reversed
Imagenette mosaics	m=4 − m=1	2,000	+0.000 [+0.000, +0.000]	null (self-nulled)

a bank that small the top-K_0 modes are mostly redundant neighbours of each other, so expansion adds no information and costs precision.

The hypothesis this kills is ours. We had pre-registered that the gain should grow with the number of targets m. Table 7 collects every test of that prediction. It is *reversed* on four corpora, confirmed on one, null on two. On 2Wiki the comparison is internal and therefore the cleanest: m=4 questions gain −0.023 [-0.044, -0.004] *less* than m=2 questions, and the m=4 stratum is a flat −0.001. What predicts a gain is hop type, not cardinality. We withdraw the cardinality prediction rather than reporting it as unsupported, because a prediction that comes out backwards on four of seven tests has been refuted, not merely unconfirmed.

Three checks, and the one that carries the weight. The expansion gain survives a disjoint question sample, a second encoder (`bge`), and banks at $\frac{1}{2}\times$ and $2\times$: HotpotQA bridge stays within +0.020 to +0.026 in all four variants, 2Wiki inference within +0.098 to +0.185. The check that matters more is the control: replace the retrieved modes with *random* bank vectors at the same K_0 and β, and the real modes win by +0.13 to +0.39 on HotpotQA and +0.16 to +0.36 on 2Wiki, in 9 of 9 cells each. The gain is therefore in *which* memories are folded into the cue, not in the act of folding something into it — which is the distinction that makes this an associative effect rather than a query-length artefact, and it is the one control we would have wanted a reviewer to demand.

One sub-claim does *not* survive. Whether the modes are kept apart (logsumexp) or averaged is encoder-dependent: the advantage of keeping them apart is +0.009 [+0.002, +0.016] on HotpotQA under `mpnet` and +0.001 [-0.004, +0.006] under `bge`. The robust finding is expansion from retrieved content; the combiner is a second-order detail, and §5.6 should be read with that in mind.

It does not transfer, which bounds the claim hard. Configurations selected on LoCoMo transfer to these corpora only in the trivial sense that LoCoMo selects K_0=0: carrying that choice over gives up the entire in-corpus gain (−0.023 on HotpotQA bridge, −0.171 on 2Wiki inference). RRF's single parameter, by contrast, transfers within 0.012 everywhere. So this is not a default anyone could ship; it is a mechanism that must be tuned on the corpus it will run on, and whose benefit is confined to one question shape. A system paper would have to detect that shape before retrieval, label-free, and we have not shown that is possible.

And a lexical baseline is still the thing to beat. Reciprocal-rank fusion is higher than expansion in absolute terms on HotpotQA (−0.017 [-0.030, -0.005] pooled), on QASPER (−0.044) and overwhelmingly on LongMemEval (−0.163 [-0.195, -0.131]). Expansion beats it on 2Wiki inference (+0.093 [+0.045, +0.140]) and compositional (+0.061), and the sign on HotpotQA flips with the encoder (+0.021 under `bge`). The honest ranking is that the two methods win in different places and that neither dominates, which is a weaker and more useful claim than either arm alone would support.

What this does to the paper's thesis. It narrows it, in the direction of being true. The claim "no departure from the dense corner pays for itself" was the shape of this paper for most of its life, and it is false: one departure pays, significantly, on a question shape we can name in advance, with a random-mode control behind it and a mechanism that makes sense. Three qualifications keep it from rescuing the rest. It is *query expansion* — T=1 with the cue folded back in — and not attractor dynamics, so it is not evidence for the energy machinery that motivated the project; it does not transfer across corpora; and on conversational memory, the setting this paper is actually about, it is null, where a lexical hybrid gains +0.156. The departure that pays is the one that uses the memory bank to *expand a query*, not the one that uses it to *complete a cue*.

And the second conversational corpus confirms the first. LongMemEval-S was the obvious replication target for LoCoMo's negative, and it reproduces it more strongly: expansion is -0.007 [-0.032, +0.018] overall, -0.049 [-0.083, -0.014] on knowledge-update questions, and the pre-registered multi-session interaction is +0.002 [-0.040, +0.045]. RRF gains +0.156 [+0.125, +0.189] overall and +0.255 on temporal-reasoning questions. Chat memory wants dates, names and exact phrases; association adds nothing on top of them. Two independent conversational corpora now say the same thing, which is as close to a settled negative as this paper gets.

5.5 Eight mechanisms that do not pay for themselves

Two findings survive from the controls in Table 9. The integrated $[1, 5, 20]$ control collapsed to exact chance ($0.108 = 1/10$) with *identical* accuracy at every occlusion level: one bad scale inside a shared trajectory poisons the entire state. The ensemble version of the same scales reaches 0.282, because there a bad scale only dilutes one vote among independent runs. This is direct evidence that the coupling between scales is real and can hurt as easily as it can fail to help.

5.6 The multi-modality result, and why it does not help

The predictor of Equation (3) returns a weighted average. A natural hypothesis, and one we committed to in a design document before testing it, is that this is a mistake: if a cue has several legitimate successors, their average is a point in embedding space where nothing is stored—a ridge between basins rather than a basin. Preserving the modes should therefore help, and should motivate a stochastic (Boltzmann) rather than deterministic relaxation (Ackley et al., 1985; Welling & Teh, 2011).

The premise is correct. Among the top three modes of $P(\cdot \mid q)$ the mean mutual cosine is 0.353, and 97% of cues have modes that mutually disagree (X24). There is genuinely a distribution for averaging to destroy.

The conclusion is wrong. With step size matched (§3), the mixture readout

$$\text{score}(m) \;=\; \text{lse}_s\big[\beta\langle \boldsymbol{\xi}_s, z_m\rangle + \log w_s\big]$$

loses to the averaged predictor at one hop, $\Delta = -0.062$ [-0.094, -0.034], and ties at two (X25). The off-manifold gap the hypothesis rested on is also much smaller than assumed: mean nearest-neighbour cosine is 0.482 for a real stored episode against 0.462 for an averaged prediction, a gap of -0.021.

The mechanism is the step size. Averaging does two things at once, and the one that matters is not the one the hypothesis named: it shrinks the displacement by a factor of 0.34 through cancellation among near-orthogonal transitions, and that shrinkage keeps the cue in control of where the prediction lands. Preserving the modes removes the shrinkage. On this data, averaging is not a failure to be fixed—it is what makes the predictor work.

A later measurement sharpens this, and the invariant turns out to be about the wrong comparison. The step-size account above predicts that where there is no displacement to shrink, the result should flip. We tested that in the modality where descent appeared to win, with a cue that is a degraded copy of two stored patterns at once (the spliced images of §6.2) and hypotheses that are the cue plus stored vectors rather than displacements, so no step size is in play. It flips, decisively: logsumexp beats averaging

Table 8: Negative results, each with its diagnosed mechanism. "Regime" indicates where the test was run. Two of these overturn claims we previously made ourselves.

Mechanism	Result	Diagnosed mechanism
Multi-step descent (G1)	no effect	Identical to one step above the β threshold—precisely what Corollary 1 predicts. Once β commits on step one there is nothing left to do (X2).
Residual peeling	0.268 vs. repulsion 0.290	Restores multi-answer recall on orthogonal synthetic patterns (0.258 → 1.000) and inverts on real embeddings: non-orthogonal golds mean projecting one out damages the others (X5, X13).
Repulsive parallel descent	+0.004, p=0.43	Survives to 146 questions but not to held-out selection. The +0.042 trend was hyperparameter overfitting (X15).
Noise-seeded parallel descent	fails at every σ	No intermediate value exists. Too little noise and all chains land in one basin; too much and they scatter uniformly, where correct answers are no likelier than any others. Isotropic noise carries no information about where the other answers are (X7).
Cosine-seeded parallel descent	exactly chance	Circular. Stored patterns are exact fixed points, so descent is a no-op; ranking the landing sites requires a signal the descent has already destroyed. Ranking them by similarity to the cue scores 1.000—but that ranking *is* cosine top-k (X7).
Multi-scale β	+0.0003 [-0.0014, +0.0021]	Expressively distinct and performance-neutral. An integrated mixture is provably not reducible to any single β (Yang et al., 2018a), and diverges from an ensemble at T=8 (63% top-1 agreement), yet changes *which* queries are answered, not how many (`exp_multibeta_scales.py`; this study predates the ledger and has no X identifier).
Per-memory / learned β	inert	The learned values (51–88) land above the leverage band, and a shuffle control with the same values randomly reassigned scores identically. Conditioning β on the query is motivated—the optimum does differ by corruption level—but the learned function converged to a near constant (β≈31 everywhere).
Predictor coupled into the energy	never exceeds external	Interpolating the routing operator inside the descent step gives 0.130 at η=0.5 and 0.154 at η=1.0—the latter merely *recovering* the external predictor, since η=1 replaces the Hopfield step entirely (X18). This was our stated novelty against prior work (Dury, 2026) and it is withdrawn.

in 20 of 21 (K_0, β) cells with intervals excluding zero, by +0.178 [+0.152, +0.205] at the cell selected on a disjoint half (X111). The gap widens with β because the averaged combiner collapses onto its heaviest mode and throws the cue away, while logsumexp degrades gracefully towards the cue.

So I4's *stated* comparison — keep the modes rather than average them — is right, and in the clean setting it is right by a wide margin. It is also irrelevant here: in the same 21 cells, **neither combiner ever beats using the cue alone** (0/21). The invariant was never the binding constraint in this experiment. Phrased as "never average modes" it is confirmed; phrased as a reason to build a multi-mode readout it is not, because the comparison that decides whether to build one is against the single-query baseline.

That last sentence had an exception, and we found it after writing it. An earlier draft of this subsection said the comparison against the single-query baseline "has never once come out in favour". §5.4

Table 9: Multi-β scales on the image task, with compute, ordered by top-1 accuracy. As in Table 1 the metric is **same-class** over ten classes — which is why the collapsed control sits at exactly 1/10 — so every row here inherits the matched baseline of §6.2 and none of them is compared against the k-NN vote. Bootstrap resamples images. Controls were pre-registered to collapse and did. Note the R@5/MRR columns: descent's shortlist is substantially worse than cosine's at every setting, which is the shortlist regression of §5.1 and Remark 1. Single β beats cosine by +0.0130 [+0.0074, +0.0186]; no mixture beats single β, and the vote of Table 1 beats all of them.

Condition	**R@1**	**R@5**	**MRR**	**matmuls**	**Δ vs. single β**
integrated mixture [32,40,48]	0.559	0.591	0.584	16	+0.0036 [-0.0033, +0.0106]
ensemble [32,40,48]	0.559	0.599	0.586	48	+0.0032 [-0.0038, +0.0103]
single beta=56	0.555	0.605	0.589	16	—
cosine	0.542	0.759	0.643	1	—
control ensemble [1,5,20]	0.282	0.296	0.303	—	collapse (predicted)
control integrated [1,5,20]	0.108	0.318	0.217	—	collapse (predicted)

is where it does: on chain-shaped multi-hop text the expanded cue beats the plain cue by up to +0.171, with selection choosing a non-null K_0 in 10 of 10 folds. The two findings are compatible and the boundary between them is informative. Here the hypotheses are the cue plus stored image vectors and the targets are the mosaic's own sources, so expansion can only dilute a cue that already contains everything needed. There the second document is not in the cue at all and can only be reached through the first, so expansion supplies information the cue does not have. What decides it is whether the retrieved modes carry anything the cue is missing — and the corpus, not the combiner, settles that.

We report this at length because it is the clearest instance in the paper of a correct premise licensing a wrong design, because the first version of the experiment was confounded by our own implementation in exactly the way §7 describes, and because the invariant then turned out to be true in a form that changes nothing.

6 The Regime Taxonomy

The results in §5 appeared to flip sign between modalities. We proposed that this is not a modality effect but a *query-type* effect, and that one sentence predicts the sign in advance. We then ran the discriminating experiments we had nominated for it (§6.2) and **none of them supported the query-type reading**. The third of them went further and removed the positive side of the comparison altogether. What survives is weaker and stated as such below: the premise predicts *losses* reliably and *wins* not at all, because after X113 there is no win left for it to predict.

Claim 1 (The completion premise)**.** Associative descent improves retrieval *only if* the cue is a degraded instance of **exactly one** stored pattern. The descent step computes a convex combination of stored patterns and moves the state onto their span, converging on a single one of them. Where the cue is not a corrupted copy of a memory, that motion destroys the cue's discriminative content with nothing to recover. Where it is a corrupted copy of *several*, the motion commits to one and discards the rest.

The converse does not hold, and we no longer have an instance of it. We originally stated this claim as "when, and only when". The "when" half is withdrawn: §6.2 satisfies the premise on text, on two image targets, and on single-target images scored by identity, and descent wins in none of the three. The premise is necessary, not sufficient — and as a sufficient condition it is not merely unproven but *contradicted on its own best case.*

Claim 1 is falsifiable in three directions and we test all three.

Satisfied, and descent still does not win. We previously placed our image study here, as the confirming case: descent's advantage over cosine was null at light corruption (+0.003, [-0.008, +0.014], n.s. at 30%) and grew monotonically to +0.021 [+0.007, +0.034] at 70%, a dose-response in the predicted direction on a pre-committed held-out split. Those numbers reproduce exactly and they do not mean what we took them to mean. The study's cues are occluded images from a disjoint split, so the cue's source image is *not in the bank*, and its metric is ten-way same-class accuracy; neither fact appeared in our earlier description of it. The

Table 10: The three regimes, the operating point each requires, and the evidence. No single fixed configuration serves more than one. †**This row no longer has a confirming measurement.** The premise satisfied on *text* yields no winning β in 45 cells; satisfied by *two* image targets at once it loses -0.245; and satisfied by single occluded images scored by identity it gives -0.003 [-0.008, +0.000] (§6.2, X113). The row is retained because the *regime* is a real one — a cue can be a degraded copy of a stored pattern — but what wins in it, in our hands, is the corner.

Regime	The cue is	What wins	Operating point
Completion†	a degraded copy of *one* stored pattern	plain cosine	T=1, $\beta \to \infty$
Composition	h steps upstream of the answer	predictor rollout	$H \geq 1$
Semantic match	a question about stored evidence	plain cosine	T=1, $\beta \to \infty$

premise was therefore never satisfied by the experiment we offered as satisfying it, and the quantity being measured was classification, not completion. §6.2 reports what happens when both defects are repaired: against a matched classifier the effect reverses, and on cues that really are degraded copies of stored patterns it is a null.

Violated, and descent loses, more as descent is applied harder. A LoCoMo question is not a corrupted conversational turn. Interpolating from pure cosine to pure descent costs accuracy at every increment, $0.336 \to 0.252$, with the optimum at the cosine boundary. A dose-response relationship in the predicted direction again, with the reduction check passing 196/196 at the boundary so that the monotonicity is a property of the method rather than an implementation artefact.

6.1 Three regimes

Claim 1 partitions agent retrieval into regimes that want mutually exclusive operating points (Table 10).

The practical consequence is sharper than a taxonomy usually is. A deployed agent meets all three regimes in one workload, so any system with a *fixed* retrieval configuration must lose two of them. This is an argument for query-conditional inference—selecting the operating point per query rather than per system—which we state as the natural next architecture. We have since built and evaluated the narrowest version of it, conditioning β alone, and **it does not work.** Setting β per query from the inverse top-two margin or the inverse score dispersion beats a fixed β=16 by +0.034 [+0.004, +0.066] and +0.067 [+0.038, +0.097]—and loses to the best fixed β, chosen on training folds only, by -0.121 [-0.149, -0.098] and -0.089 [-0.121, -0.061] (X65-R). The apparent gain was the reference point: β=16 scores 0.241 where the fold-selected β=32 scores 0.397, so conditioning the knob recovers part of a handicap rather than adding anything. A sign-reversed control ($\beta \propto$ margin) costs -0.390 [-0.433, -0.348], so the rules do read the signal they were designed to read; they simply cannot spend it.

That is a result about one knob, not about the taxonomy, which asks for selection among *regimes*—and that remains unbuilt. But it is a pointed one, because this is the component we flagged as most exposed to the selection error of §7: "choose the budget that maximises the score" is test-set tuning with extra steps unless the selector is fitted and frozen on training data. What went wrong first was not the selector but the *baseline* it was scored against.

6.2 The discriminating test, now run

An earlier version of this section nominated a text occlusion study as the cheapest discriminating test, predicted that descent would regain its advantage, and committed in advance to the alternative reading: "if it does not, the premise is wrong and the modality explanation is right." We flagged it as the experiment most likely to falsify our own central claim. It has now been run, pre-registered, and **it falsified the prediction** (X110).

Design. LoCoMo corpus and encoder held fixed; only the query type changes. The cue is a stored turn with a contiguous span of its words removed — a span, because the image study blacks out a square *patch*

— at the same three severities the image study used. The gold item is that turn, so the premise holds by construction and the task is single-target. β was selected on a disjoint half of the conversations and used once. 5,547 eligible turns; the identity control at zero masking returns recall@1 = 1.000, and cosine has ample headroom ($0.934 \to 0.721 \to 0.365$ across the three levels).

Result. Descent never wins. At the selected β the held-out Δ recall@1 is -0.0007 [-0.0013, +0.0000] at 70% masking, with no dose-response. Widening the search to fifteen β values from 1 to 1024 crossed with the three severities gives 0 **of** 45 **cells with a positive** Δ recall@1**, and** 0 **of** 45 **for** recall@5. The harm falls monotonically in β and asymptotes at zero — which is Corollary 1 rather than a band: the best available configuration is the one that reduces to cosine. On images a β existed where descent beat *cosine* — though not, as the next subsection shows, where it beat a classifier — while on text there is no such β at all.

And the shortlist cost does not vanish. Even at $\beta = 1024$, where rank 1 is cosine's by the reduction, Δ recall@5 stays at -0.029, -0.111 and -0.144 across the three severities. That is Remark 1 measured at the limit on 5,547 text probes: ranks $2..k$ read off the converged state order by proximity to $z_{i^\star}$ forever, so the regression is structural and not a tuning artefact.

The third direction: more than one target. The premise says "a degraded instance of a stored pattern". If the cue is a degraded instance of *two*, the descent must still converge on one. We built that cue in the modality where descent appeared to win — splice the left half of one stored image onto the right half of another, always across classes, so the gold set is both source images — and descent loses heavily: Δ gold@5 $= -0.245$ [-0.275, -0.215] against cosine on held-out cues, with β again selected on a disjoint half (X111). Single-target is not a detail of the premise; it is the premise.

The fourth direction: auditing the win itself (X113)

Both tests above are of the form "satisfy the premise somewhere new". The remaining direction is to ask whether the one place we claimed it *was* satisfied really was. It was not, in two independent ways, and we found this by reading our own scoring code rather than by running anything.

What the image study actually measured. Its bank is 400 images drawn from the `train` split and its cues are occluded images from the `val` split, so **the cue's source image is never in the bank**: the cue is a degraded instance of an *unstored* pattern that shares a class label with 40 stored ones. And a retrieval is scored correct when the returned memory shares the query's class, over ten classes — **the metric is classification accuracy, not identification**. Chance is 0.100, which is the number we had already reported as the collapse floor without remarking that it is also the chance rate. Neither fact was stated in our earlier description of the experiment, and together they mean the study never tested the premise it was offered as satisfying.

The mechanism this leaves. At the selected β=40 the first descent step has an effective support of 12.7 patterns (median; mean 15.9, p_{90} 32.2) out of 400, with 48% of the blend mass on the cue's own class. On a class metric, averaging over a same-class neighbourhood of that size is a *soft k-nearest-neighbour class vote*, which is a standard classifier containing no energy, no iteration and no attractor. Cosine top-1 is a 1-NN classifier and gets no such ensemble, so our published comparison was between a hard classifier and a soft one.

The matched baseline reverses the result. We added the explicit vote: score each class by $\sum_{i \in \mathcal{N}_k(q),\, c_i = c} \mathrm{softmax}(\tau \langle z_i, q \rangle)$ and predict the argmax, with k and τ selected on the same tune half that chose β, from grids declared in advance ($k \in \{1, 3, 5, 10, 20, 40\}$, $\tau \in \{4, \ldots, 64\}$; k=1 reproduces cosine exactly, an internal null that passes at every τ). Selection chose k=20 — the same order as the 12.7 patterns descent blends implicitly. On the held-out half (Table 1) the vote scores 0.602 against descent's 0.566 and cosine's 0.553: descent is **worse than the matched baseline at every occlusion level**, pooled -0.036 [-0.043, -0.028]. Descent's +0.014 over cosine is real and is roughly a quarter of what the plain classifier gets for free.

And the premise's own best case is a null. Finally we built the experiment our prose had always described: occlude the *bank* images themselves, so the cue is genuinely a degraded copy of a stored pattern, and score **same-image** recall@1. The identity control passes exactly (cosine recall@1 = 1.0000 at 0% occlusion). On the held-out half (Table 2) descent gives −0.003 [-0.008, +0.000] against cosine, and β selection on the tune half chose 80, the top of the grid — Corollary 1 again, selection walking toward the corner. Same-class accuracy on the identical cues is also null (−0.002). Where the premise is satisfied exactly, in the modality where we thought descent won, and with the metric that matches the word "completion", there is nothing there.

An earlier run had already measured this, and we state exactly how much that adds (X210). Two days before the audit above, a pre-registered run in the six-corpus programme of §5.4 added a k-NN class-vote control to the image arm for its own reasons and reported descent −0.0313 [-0.0397, -0.0227] against the vote. We are deliberate about what that corroborates. **It is not an independent replication of the image result**: X210 and X113A evaluate the same 5,847 cues over the same 1,949 source images against the same bank, and their descent-minus-cosine figures agree to sixteen decimal places because they are the same measurement. What differs is only the vote's grid and the half that selected it, which is why the contrast against the vote moves from −0.0313 at k=10 to −0.0356 at k=20. Two runs of one experiment are one experiment.

What X210 does add is a modality X113 never ran. The same three-way comparison on YAHOO ANSWERS with spans of the stored text deleted gives descent +0.003 [-0.000, +0.006] over cosine — not significant, p=0.09 — while the k-NN vote gains +0.104 [+0.088, +0.120]; with `[MASK]` substituted for deletion the numbers are +0.002 and +0.111. So on text the energy path does nothing measurable, and the class smoothing it partially recovers on images is available in full, and far more cheaply, from a plain neighbourhood vote. That is a second modality agreeing with the reversal, and it is the part of X210 we weight.

What we conclude. The query-type reading is not supported, and we withdraw it rather than defend it. The completion premise still predicts every *loss* we have measured — text question→evidence, text masked turns, two-target image splices, and now single-target image identification — and it no longer has a *win* to predict. The taxonomy's three regimes stand as a description of where descent loses, which is what the evidence supports, and the claim that one sentence predicts the sign in advance does not.

What does not follow is that nothing outside the corner ever pays. §5.4 reports one mechanism that does, on a question shape we can name before running it, and the distinction between the two findings is the useful part: *completing a cue* from the memory bank does not work anywhere we have looked, and *expanding a query* with retrieved memories works where the next piece of evidence is reachable only from the retrieved evidence. Both are departures from (T=1, $\beta \to \infty$, b=0, R=I). Only the second one pays, and it is not the one that motivated this paper.

We state plainly what this costs the paper. X113 removes the only positive result we had for the energy machinery itself, and it is a result we had published a pre-registered, split-clean, bootstrapped confirmation of. What it does not touch is §5.3: the nonparametric predictor does not read an answer off the landscape and its one-hop advantage is unaffected by anything here. The paper's single surviving positive is the mechanism that is not associative retrieval.

7 What the Protocol Was Worth

This section reports seven errors we made, and for each one the magnitude of the false result it produced in our own data. We include it because the errors are not exotic—each is a default that a reasonable person reaches for—and because their cost here is large enough to flip conclusions. The point is not confession. It is that for this class of experiment, the protocol is not a hygiene matter around the result; it *is* the result, and we can put numbers on that.

Three observations we think generalise beyond this paper.

The errors are not symmetric in how they feel. Two of the seven produced false *negatives*, and those were by far the harder to find, because a null result invites no scrutiny. The dimensionless-hyperparameter bug reported a working mechanism as 0.016 when its true value was 0.277; it was found only because the number was suspiciously round. A field that publishes negative results—which we are arguing it should—inherits an obligation to audit them as hard as positives.

Self-replication caught more than review would have. The two-hop result in §5.3 and the mode-averaging result in §5.6 were both contradicted by reimplementation, not by inspection. In both cases the original code had been read, reviewed and believed. We now treat "has this been written twice, independently" as a distinct and stronger property than "has this been reviewed", and we report which of our claims have it: one does.

An uncommitted script is not a result. The uncommitted-script error is different in kind from the others: it cannot be corrected, only recorded. Both drivers that were lost have now been rewritten, and rewriting them is what revealed that one of the two results did not reproduce. We state the rule the episode produces, because it is cheap and we did not follow it: a number does not enter the record until the code that produced it is committed.

A protocol cannot check that the baseline was allowed to do its best. The seventh error is the one we would most want a reader to take away, because every safeguard we had was already in place when we made it. Our image result had a disjoint tuning split, a pre-committed comparison, a paired bootstrap and a published grid, and it was still an artifact — because all of those check *how* two arms are compared and none of them check *which* two arms. The method was a soft classifier and the baseline was a hard one; the metric was class identity, which is not what "pattern completion" means. The rule we now follow is stated in our ledger and is not a statistical one: before reporting a delta, write down what the other arm was allowed to do, and check it was allowed to do its best. Where a comparison has two arms, print a third.

The reduction is what made the audit possible. Because the baseline is a configuration of the same code (Proposition 1), four of the seven errors above were detectable as a *violation of a known equivalence*—the method failing to reproduce cosine where it provably must. An architecture in which every component has a null setting that reduces to the component below it converts a class of silent bugs into failing assertions. We regard this as the main transferable engineering claim of the paper, separate from any empirical finding: build the baseline as a corner of the method, and assert the reduction in continuous integration.

8 Limitations

Two public benchmarks, one task shape. The one-hop positive replicates on LoCoMo (Maharana et al., 2024) and on MSC (Xu et al., 2022) at a bank five times the size (§5.3), which discharges the external-validity concern for that result about as well as two corpora can. It does not discharge it generally, and the reason is the *probe*, not the corpus. Cue a turn, retrieve the turn that followed is one operationalisation of temporal association, and both corpora are speaker-alternating dialogue, so both give next-turn prediction a regularity that a single agent's episode stream may not share. Replicating on a second corpus of the same shape cannot detect a bias that belongs to the shape. What this claim still lacks is a corpus where the stream is one agent's own actions, and we know of no public one.

Association barely composes, and we do not know why. Two hops is null on the episodic corpus and on LoCoMo, and positive but 18× smaller than the withdrawn claim on MSC (§5.3). We attribute it to compounding prediction error exceeding the compositional signal, but that is an interpretation rather than a measurement: we have not isolated the error accumulated per step, and the small residual Recall@5 effect on LoCoMo is unexplained. A rollout-depth sweep with per-step error attribution is the experiment this calls for.

The retrieval task is not the downstream task, and the gap is now measured. Every number here is retrieval quality. We have since built the cheapest honest proxy for the downstream question —

normalised substring coverage of the gold answer inside the concatenated top-k turns, on LoCoMo's own annotated answers, deterministic and API-free — and **a large retrieval gain does not survive into it** (X35). A reciprocal-rank-fusion baseline that beats dense cosine by +0.124 on evidence recall@10 carries that into +0.007 [-0.013, +0.027] of answer coverage at k=10: indistinguishable from zero. Coverage is a weak proxy and we call it coverage rather than accuracy for that reason, but it moves in the direction that should be easiest to move, and it did not. Whether *any* of these retrieval differences change what an agent does remains untested, and this result makes us less optimistic rather than more.

Our pseudo-relevance-feedback baseline never helps, anywhere, and we cannot yet say why. Rocchio feedback is the prior art a reader names first, and in our hands it is selected straight to its null (γ=0) on every corpus and loses whenever it is forced to do something. We therefore ran it on SciFact, a standard ad-hoc retrieval benchmark where feedback methods are well calibrated: 0 of 16 swept cells beat cosine there either (X104). Two readings remain open — that a strong dense baseline leaves classical PRF nothing to add, which the literature on dense feedback would support, or that our PRF is simply weak. Until that is settled, we do not claim "the prior art does not help either", and any comparison in this paper against Rocchio should be read with that reservation.

Scale. Banks run from 47 patterns (QASPER, one paper) to 30,320 (MSC), with encoders of 512, 768 and — for the cross-modal arm — CLIP's joint space. Both ends of that range matter and both are far from the target: the 47-paragraph banks are where expansion inverts and starts to hurt, and the 30,320-turn bank is still two orders of magnitude short of what a long-lived agent accumulates. The image advantage is explicitly capacity-dependent—cosine overtakes descent past $N \approx 800$ at fixed β—so nothing here licenses a claim at the 10^6-memory scale a long-lived agent would reach. Whether the crossover is a β-scaling artefact is unresolved. It was the most important unanswered question about the completion result until X113 removed the result; we retain it because the capacity behaviour of descent is of independent interest.

The latency cost is real, and larger than the matmul count suggests. Descent at T=8 issues 16 matmuls against cosine's 1, but measured wall-clock is 21× (0.0502 ms vs. 0.0024 ms per query), for a pooled +0.014 same-class accuracy that a k-NN class vote exceeds at a fraction of the cost (§6.2). Multi-β variants cost a further 1.6–2.9× for an effect indistinguishable from zero. We do not think this trade is attractive in deployment and say so.

The shortlist regression. Descent improves rank 1 and degrades ranks 2–5 (0.605 vs. 0.759 Recall@5). For an agent that pastes five memories into context, which is the standard usage, this is plausibly a net loss.

The image study was not a completion study, and we did not notice for most of this project. Its metric was same-class accuracy and its cues were drawn from a split disjoint from the bank, so the premise it was offered as confirming was never satisfied by it (§6.2, X113). We report this as a limitation of our own process, not only of the result: the experiment had a pre-committed split, a declared grid and a paired bootstrap, and none of that machinery can detect a baseline that was never allowed to do its best, or a metric whose name does not match the operation it scores. Readers weighing the rest of this paper should know that the defect was found by re-reading our scoring code, which is the one audit we had not systematised.

The one positive departure is not deployable, and we say so rather than rounding it up. Expansion on chain-shaped multi-hop questions (§5.4) is the only mechanism in this paper that beats the dense corner on its own terms. Three things bound it. It requires tuning on the corpus it will serve — configurations selected on LoCoMo give up the entire gain, where RRF's single parameter transfers within 0.012. It requires knowing the question's shape in advance, and we have not shown a label-free signal that separates a chain question from a comparison question before retrieval; the gain and the harm live in different strata of the *same* corpus, so a system cannot collect the one without risking the other. And it is null on both conversational corpora, which is the setting this paper is nominally about. A reader looking for the deployable recommendation in this paper will not find it here either.

Two strata carry more weight than their size supports. The cardinality table (Table 7) rests in two places on a 23-query stratum: SciFact multi-relevant and NFCorpus single-relevant. One of those two is the single confirmation of the prediction, so the one piece of evidence *for* our own hypothesis is also among the weakest in the table. We have left both rows in rather than dropping them, and flagged them in the caption, because dropping the inconvenient-but-weak row and keeping the convenient-but-weak one is precisely how a table acquires a thumb on the scale.

An experimental record can fork, and ours did. Two programmes in this project ran concurrently and both numbered their experiments `X100`–`X112`, so the same identifier names different experiments depending on which record a reader consults, and a merge of the two documentation files resolved in favour of one — leaving a committed documentation file that described seven completed corpora as "not run" while their result files sat in the repository. We renumbered the second programme to `X200`–`X212` and the ledger now carries both phases. We report it because it is the same failure as the others in §7, one level up: not a wrong number, but a record that disagreed with the data on disk, and no test can catch that — the results were correct, reproducible and committed the whole time, and only the prose describing them was wrong.

Biological framing is analogy. We use the words basin, attractor and consolidation because they are the established vocabulary of the formalism. We make no claim of correspondence with neural mechanism, and no result here depends on one.

The remaining axes have now been tested, and none of them helps. An earlier version of this section listed G2 and the controller as unevaluated. Both have since been measured, on a multi-target variant of the same LoCoMo corpus—retrieve the full set of annotated evidence turns for each of 1,977 questions—with the conversation as the resampling cluster. Ten clusters floor a blocked-permutation p at $2/2^{10} = 0.002$, so we give intervals and not p-values.

G2 (strength-modulated basin depth) hurts under every prior we tried—recency, centrality, session size, turn length—and hurts monotonically in the bias strength. The gentlest setting is indistinguishable from no bias at all (-0.004 [-0.035, +0.030]) and the strongest costs -0.495 [-0.533, -0.459] (X62). **G3 with a *fitted* routing operator** reproduces the X18 verdict by an independent route: diagonal routing costs -0.029 [-0.046, -0.013] at its mildest and dense routing -0.471 [-0.512, -0.424] at its strongest, so $R{=}I$—the corner—is the best routing we have been able to find (X63). The lifecycle operations built on G2, decay and consolidation, are still implemented and unevaluated; that part of the gap stands.

A single encoder per modality. Proposition 1 is encoder-independent, but the β band is not, and we measured it on one text encoder and one vision backbone. The claim that the band must be re-measured per encoder is supported by two points (synthetic and `all-mpnet-base-v2`), which establishes that it varies but not how it varies.

9 Conclusion

We set out to build a better memory for autonomous agents on associative-memory foundations. That thesis is refuted here by our own measurements, at adequate power, with the mechanism identified: every departure from the dense-retrieval corner that changes only how an answer is read off the energy landscape fails on agent text, because none of them changes the landscape. The one departure that survives changes the *cue* instead, and it is query expansion rather than attractor dynamics.

What replaced it is narrower and better supported. Dense retrieval is exactly the $(T{=}1, \beta \to \infty, b{=}0, R{=}I)$ corner of the modern Hopfield update — exactly so at $k{=}1$, and at $k > 1$ with the shortlist read off the cue — which makes the standard baseline a configuration of the method under test rather than a rival outside it, and converts an ill-posed question into a finite one. Within that frame, one side of the sign is predictable before the experiment: associative descent hurts monotonically when the cue is not a degraded instance of a stored pattern, and that half has held everywhere we have looked. The other half has no surviving instance. We nominated three tests against ourselves and all three came back negative: satisfying the premise on masked text gives no winning β in 45 cells, two image targets instead of one costs -0.245, and

the single-target image result we had treated as the confirming case turns out to be a ten-way classification measured on cues whose source was never stored, which an explicit k-NN class vote performs better than descent does (-0.036, [-0.043, -0.028]). On the premise's own best case — occluded copies of stored images, scored by identity — descent is -0.003 [-0.008, $+0.000$]. The premise is therefore necessary and not sufficient, and we withdraw the image win rather than keep it under a weaker description (§6.2). Of eight mechanisms beyond the corner, one survives—a nonparametric predictor over the agent's own observed transitions, which retrieves by succession rather than resemblance. It beats cosine on one-hop association in four measurements across three corpora, including +0.043 Recall@1 [$+0.036, +0.050$] on LoCoMo against a pooled bank of 5,882 turns and +0.016 [$+0.014, +0.018$] on MSC against 30,320. The same experiments bound it: at two hops the effect is -0.002 on the first corpus and +0.005 on the second, one eighteenth of the +0.185 two-hop result of our own that we withdraw. Association helps one step ahead and essentially stops there.

One thing beats the corner, and it is not the thing we were looking for. We could not tell from LoCoMo alone whether query expansion fails as a mechanism or fails on that corpus, so we ran six more. It fails on the corpus. Fold a few retrieved memories back into the cue and retrieve again, and on multi-hop questions whose second document is reachable only *from the first* the gain is +0.023 [$+0.010, +0.037$] and +0.171 [$+0.118, +0.219$]; on questions in the same corpora whose documents are both reachable from the question, selection declines to expand in 10 of 10 folds and the method correctly does nothing. Random modes lose to real ones by +0.13 to +0.39, so what pays is which memories enter the cue. Our own pre-registered explanation for when this should happen—more targets—comes out reversed on four corpora, and the predictor that survives is the shape of the hop. We hold the claim at arm's length nonetheless: it needs per-corpus tuning, it needs a question shape we cannot yet detect without labels, and on the two conversational corpora that motivate this paper it is null, where a lexical hybrid gains +0.156. It is the one place where the thesis we began with was too pessimistic, and it is still not a better agent memory (§5.4).

We also report what the protocol was worth, in the only currency that makes the argument: seven errors, each a reasonable default, each worth a conclusion. Two produced false negatives, which were materially harder to detect than the false positives. Three of our own recorded claims were overturned by reimplementation rather than by review—and the one that had been our headline was settled only by the public-benchmark run that external validity demanded anyway. The seventh is the one a reader should weigh most heavily, because it survived every check we had: an *unmatched baseline*. The image comparison was between a 1-nearest-neighbour classifier and a soft-k-nearest-neighbour one, on a class metric, and no amount of split hygiene, pre-registration or bootstrap discipline detects that. Only naming the operation the method actually performs does.

The constructive proposal we take from this is structural rather than empirical. Build the baseline as a corner of the method, give every added component a null setting that reproduces the configuration beneath it, and assert those equivalences in continuous integration. Four of the seven errors above surfaced as violations of a known equivalence rather than as anomalous numbers. In a literature where the operator under test and the operator it is compared against coincide at the default operating point, that is not a convenience. It is the difference between measuring a method and measuring a bug.

Author Contributions

Pushkal Gupta led the project and is responsible for the substance of this paper. He formulated the reduction (Proposition 1) and its corollary, derived the proof in Appendix A, designed and implemented the energy core, the nonparametric predictor, the mixture readout, and the evaluation harness, and ran and analysed every experiment reported here, including the LoCoMo and MSC external-validity studies, the six-corpus expansion study of §5.4, and the protocol-error audit of §7. He built the experiment ledger and the generate-never-type table pipeline, and wrote the paper.

Rajit Agrawal maintained the bibliography and the citation audit, and contributed to the experimental-protocol documentation.

Anshveer Singh contributed the initial image-modality test harness and assisted with running the occlusion experiments.

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References

David H. Ackley, Geoffrey E. Hinton, and Terrence J. Sejnowski. A learning algorithm for boltzmann machines. *Cognitive Science*, 9(1):147–169, 1985. doi: 10.1016/S0364-0213(85)80012-4.

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15619–15629, 2023. doi: 10.1109/CVPR52729.2023.01499.

A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 758–759, 2009. doi: 10.1145/1571941.1572114.

Pradeep Dasigi, Kyle Lo, Iz Beltagy, Arman Cohan, Noah A. Smith, and Matt Gardner. A dataset of information-seeking questions and answers anchored in research papers. In *Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 4599–4610, 2021. doi: 10.18653/v1/2021.naacl-main.365.

Jason Dury. Predictive associative memory: Retrieval beyond similarity through temporal co-occurrence. *arXiv preprint arXiv:2602.11322*, 2026.

Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of bert, elmo, and gpt-2 embeddings. In *Conference on Empirical Methods in Natural Language Processing*, pp. 55–65, 2019. doi: 10.18653/v1/D19-1006.

Jun Gao, Di He, Xu Tan, Tao Qin, Liwei Wang, and Tie-Yan Liu. Representation degeneration problem in training natural language generation models. In *International Conference on Learning Representations*, 2019.

Taosha Guo and Fabio Pasqualetti. Learnable sequential memory in coupled oscillator networks. *arXiv preprint arXiv:2607.18439*, 2026.

Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. Hipporag: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems*, pp. 59532–59569, 2024.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016. doi: 10.1109/CVPR.2016.90.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. In *International Conference on Computational Linguistics*, pp. 6609–6625, 2020. doi: 10.18653/v1/2020.coling-main.580.

John J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982. doi: 10.1073/pnas.79.8.2554.

Dmitry Krotov and John J. Hopfield. Dense associative memory for pattern recognition. In *Advances in Neural Information Processing Systems*, 2016.

Yann LeCun. A path towards autonomous machine intelligence. Position paper, Version 0.9.2, OpenReview, June 2022.

Chong Li, Taiping Zeng, Xiangyang Xue, and Jianfeng Feng. Dynamic manifold hopfield networks for context-dependent associative memory. *arXiv preprint arXiv:2506.01303*, 2025.

Qiang Liu and Dilin Wang. Stein variational gradient descent: A general purpose bayesian inference algorithm. In *Advances in Neural Information Processing Systems*, 2016.

Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of llm agents. In *Annual Meeting of the Association for Computational Linguistics*, pp. 13851–13870, 2024. doi: 10.18653/v1/2024.acl-long.747.

Stéphane G. Mallat and Zhifeng Zhang. Matching pursuits with time-frequency dictionaries. *IEEE Transactions on Signal Processing*, 41(12):3397–3415, 1993. doi: 10.1109/78.258082.

Beren Millidge, Tommaso Salvatori, Yuhang Song, Thomas Lukasiewicz, and Rafal Bogacz. Universal hopfield networks: A general framework for single-shot associative memory models. In *International Conference on Machine Learning*, 2022.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, 2021.

Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021.

Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Conference on Empirical Methods in Natural Language Processing*, pp. 3980–3990, 2019. doi: 10.18653/v1/D19-1410.

Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: Bm25 and beyond. *Foundations and Trends in Information Retrieval*, 4(1–2):1–174, 2009. doi: 10.1561/1500000019.

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. BEIR: A heterogenous benchmark for zero-shot evaluation of information retrieval models. In *Conference on Neural Information Processing Systems, Datasets and Benchmarks Track*, 2021.

Max Welling and Yee Whye Teh. Bayesian learning via stochastic gradient langevin dynamics. In Lise Getoor and Tobias Scheffer (eds.), *Proceedings of the 28th International Conference on Machine Learning (ICML-11)*, pp. 681–688, New York, NY, USA, 2011. ACM. ISBN 978-1-4503-0619-5.

Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. Longmemeval: Benchmarking chat assistants on long-term interactive memory. In *International Conference on Learning Representations*, 2025.

Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. C-pack: Packed resources for general chinese embeddings. In *International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 641–649, 2024. doi: 10.1145/3626772.3657878.

Jing Xu, Arthur Szlam, and Jason Weston. Beyond goldfish memory: Long-term open-domain conversation. In *Annual Meeting of the Association for Computational Linguistics*, pp. 5180–5197, 2022. doi: 10.18653/v1/2022.acl-long.356.

Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-mem: Agentic memory for llm agents. In *Advances in Neural Information Processing Systems*, pp. 20004–20031, 2025.

Zhilin Yang, Zihang Dai, Ruslan Salakhutdinov, and William W. Cohen. Breaking the softmax bottleneck: A high-rank rnn language model. In *International Conference on Learning Representations*, 2018a.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Conference on Empirical Methods in Natural Language Processing*, pp. 2369–2380, 2018b. doi: 10.18653/v1/D18-1259.

Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. Memorybank: Enhancing large language models with long-term memory. In *AAAI Conference on Artificial Intelligence*, volume 38, pp. 19724–19731, 2024. doi: 10.1609/aaai.v38i17.29946.

Jinchang Zhu, Jindong Li, Cheng Zhang, Jiahong Liu, and Menglin Yang. Hela-mem: Hebbian learning and associative memory for llm agents. *Proceedings of ACL*, 2026.

A Proof of Proposition 1 and Corollary 1

Throughout, $X = [z_1 \cdots z_N] \in \mathbb{R}^{d\times N}$ with $\|z_i\| = 1$, the cue satisfies $\|q\| = 1$, and we set $b = 0$, $\eta = 0$, $T = 1$. Write $s_i = \langle z_i, q\rangle \in [-1, 1]$ for the logits, $p(\beta) = \mathrm{softmax}(\beta s) \in \Delta^{N-1}$, and $\boldsymbol{\xi}(\beta) = Xp(\beta)$ for the unnormalised state after one update.

A.1 The unique-maximiser case

Lemma 1 (Softmax concentration)**.** *Let $i^\star = \arg\max_i s_i$ be unique and let $\delta = s_{i^\star} - \max_{i\neq i^\star} s_i > 0$ be the top-two gap. Then*

$$1 - p_{i^\star}(\beta) \;\le\; (N-1)\, e^{-\beta\delta}, \qquad \textit{and} \qquad \sum_{i\neq i^\star} p_i(\beta) \;\le\; (N-1)\, e^{-\beta\delta}.$$

Proof. $p_{i^\star}(\beta)^{-1} = 1 + \sum_{i\neq i^\star} e^{-\beta(s_{i^\star} - s_i)} \le 1 + (N-1)e^{-\beta\delta}$, so $p_{i^\star} \ge (1 + (N-1)e^{-\beta\delta})^{-1} \ge 1 - (N-1)e^{-\beta\delta}$, using $(1+u)^{-1} \ge 1 - u$ for $u \ge 0$. The second bound is $1 - p_{i^\star}$. □

Lemma 2 (State convergence with rate)**.** *Under the hypotheses of Lemma 1, $\big\|\boldsymbol{\xi}(\beta) - z_{i^\star}\big\| \;\le\; 2(N-1)\, e^{-\beta\delta}$. Consequently $\boldsymbol{\xi}(\beta) \to z_{i^\star}$ as $\beta \to \infty$, and since $\|z_{i^\star}\| = 1$ the projection Π is continuous at the limit and $\Pi[\boldsymbol{\xi}(\beta)] \to z_{i^\star}$.*

Proof. Write $\varepsilon = \sum_{i\neq i^\star} p_i = 1 - p_{i^\star}$. Then $\boldsymbol{\xi} - z_{i^\star} = (p_{i^\star} - 1)z_{i^\star} + \sum_{i\neq i^\star} p_i z_i$, so by the triangle inequality and $\|z_i\| = 1$, $\|\boldsymbol{\xi} - z_{i^\star}\| \le \varepsilon + \varepsilon = 2\varepsilon$. Apply Lemma 1. For the last statement, $\|\boldsymbol{\xi}\| \ge \|z_{i^\star}\| - 2\varepsilon = 1 - 2\varepsilon > 0$ for β large, so Π is well defined and Lipschitz in a neighbourhood of $z_{i^\star}$. □

This proves the state half of Proposition 1, with an explicit $O(e^{-\beta\delta})$ rate governed by the top-two similarity gap. Note the rate depends on δ, not on N except logarithmically: $\beta \ge \delta^{-1}\log\big((N-1)/\epsilon\big)$ suffices for $\|\boldsymbol{\xi} - z_{i^\star}\| \le 2\epsilon$.

A.2 The $k > 1$ case: ranking, not state

Proposition 1 as used in this paper is a statement about the *top-k set*, which requires a separate argument because the state converges to a single pattern and the ranking is read off that state. Let π_β be the ordering of $\{1..N\}$ by $\langle z_i, \Pi[\boldsymbol{\xi}(\beta)]\rangle$ descending, and let $\pi_{\cos}$ be the ordering by $\langle z_i, q\rangle$ descending.

Lemma 3 (Rank-1 agreement)**.** *Under the hypotheses of Lemma 1, for β sufficiently large $\pi_\beta(1) = \pi_{\cos}(1) = i^\star$.*

Proof. By Lemma 2, $\langle z_{i^\star}, \Pi[\boldsymbol{\xi}]\rangle \to 1$ while for $j \neq i^\star$, $\langle z_j, \Pi[\boldsymbol{\xi}]\rangle \to \langle z_j, z_{i^\star}\rangle \leq 1$ with equality only if $z_j = z_{i^\star}$, excluded by distinctness of the patterns. Choosing β past the point where the $O(e^{-\beta\delta})$ perturbation is smaller than half the gap $1 - \max_{j\neq i^\star}\langle z_j, z_{i^\star}\rangle$ gives the claim. □

Remark 1 (Ranks 2..k do not agree, and this is a result, not a gap). For $k > 1$ the two orderings are *not* equal in the limit. Ranks 2..k of π_β order the patterns by proximity to $z_{i^\star}$, whereas $\pi_{\cos}$ orders them by proximity to q. The reduction therefore holds exactly at $k = 1$ and as a statement about the retrieved *set* only when q is close enough to $z_{i^\star}$ that the two neighbourhoods coincide.

This is not a technical caveat; it is the mechanism behind the empirical shortlist regression in §5.1. Descent improves top-1 while degrading recall@5 (0.605 vs. cosine's 0.759; both class-level, as everywhere in the image study — `exp_multibeta_scales.py` computes them that way and labels them so) precisely because the converged state has forgotten the cue. Our implementation of the dense baseline therefore reads the top-k from q directly, which is both the standard definition and the configuration that `tests/test_reduction.py` verifies; a baseline reading ranks 2..k off the converged state would be a weaker baseline and would flatter the method.

A.3 Ties

If the maximiser is not unique, let $\mathcal{M} = \arg\max_i s_i$ with $|\mathcal{M}| = m \geq 2$. Then $p_i(\beta) \to 1/m$ for $i \in \mathcal{M}$ and 0 otherwise, so $\boldsymbol{\xi}(\beta) \to \frac{1}{m}\sum_{i\in\mathcal{M}} z_i$, the centroid of the tied patterns, and Π maps it to the normalised centroid provided that sum is non-zero. The retrieved top-1 is then whichever tied pattern is nearest that centroid, which is an arbitrary tie-break — exactly as it is for exact cosine retrieval.

We record this case because it is the low-β centroid-collapse failure of §3.4 in miniature. At finite, small β the softmax is flat not because of exact ties but because $\beta\delta \ll 1$ for many pairs, and the state lands near the centroid of a large effective set. At β=16 on a 768-dimensional sentence encoder the effective set size $\exp(H(p))$ is 377 patterns. The tie case and the collapse case are the same phenomenon at different scales, which is why a single lower bound on β describes both.

A.4 Proof of Corollary 1

Each z_i is a fixed point of the update in the $\beta \to \infty$ limit: if $\boldsymbol{\xi}^{(t)} = z_j$ then $s = X^\top z_j$ is maximised uniquely at j (patterns distinct, unit norm), so by Lemma 2 the next state is z_j again. Combined with Lemma 2 applied at $t = 0$, the trajectory reaches $z_{i^\star}$ after one step and remains there, so $T > 1$ is observationally equivalent to $T = 1$, which is cosine retrieval by Proposition 1.

Scope of the empirical claim. Corollary 1 is exact only in the limit. The empirical statement in §3.4 — that a comparison run at high β compares cosine against cosine — is therefore an *approximation claim*, and we state its evidential basis precisely rather than leaning on the theorem. On the held-out image task at β=512, descent returns *the same memory as cosine on* 99.4% *of queries* (same-class accuracy 0.4986 against the cosine baseline's 0.5000). Decision agreement is the right statistic and accuracy is not: two retrievers can score identically while disagreeing about what they return, and the corollary is a claim about the choice. The theorem explains *why* the gap closes and gives the rate at which it closes; the measurement establishes that at the β values this literature actually uses, it has closed. Neither alone supports the claim.

A.5 Numerical verification

`tests/test_reduction.py` instantiates Equation (2) with T=1, β=10^6, b=0, η=0 and asserts top-1 agreement with an independent exact-cosine implementation, in two settings. On isotropic synthetic Gaussians (d=64, N=500) it agrees on 1000/1000 random queries. On real sentence embeddings (d=768, `all-mpnet-base-v2`) it agrees on 838/838 perturbed-episode queries.

The second setting is the one that carries weight, and we separate them deliberately. Random high-dimensional vectors are near-orthogonal, so the top-two gap δ of Lemma 1 is large and concentration is trivial. Real embeddings are anisotropic — unrelated pairs sit at cosine 0.258, not 0 — so δ is substantially

smaller and the reduction is a materially stronger claim there. Both runs write their agreement counts to `experiments/results/reduction_check.json`; the test gates continuous integration on every commit. We use β=10^6 rather than a symbolic limit because that is the operating point the claim is about; Lemma 1 shows $\beta\delta \gg \log N$ is the sufficient condition and for $N \leq 10^4$, $\delta \geq 10^{-3}$ it is met with large margin.

B Hyperparameter grids, splits, and tuning budget

The β grid in row 1 is swept densely through the transition region and coarsely outside it, and `sweep_beta_transition.py` records, for every value, both same-class accuracy and the top-1 decision agreement with cosine plotted in Figure 1 (`experiments/results/beta_band.json`).

Cost. Measured wall-clock on the hardware of Appendix G, per query, on the held-out image split: cosine 0.0024 ms (1 matmul), single β at T=8 0.0502 ms (16 matmuls), integrated multi-β mixture 0.0810 ms (16 matmuls), and the m-way ensemble 0.1434 ms (48 matmuls). The ratio that matters for deployment is the first against the second: a 21× wall-clock cost for +0.013 same-class accuracy over cosine — a margin the k-NN class vote of §6.2 exceeds for one extra argpartition. The mixture variants add a further 1.6–2.9× for an effect indistinguishable from zero (Table 9).

We report matmul counts alongside wall-clock because the two diverge: the integrated mixture and the single-β run issue the same 16 matmuls but differ 1.7× in wall-clock, since mixing m softmaxes per step is $O(mN)$ arithmetic outside the matmuls – 1.6× the wall-clock for the same matmul count. A matmul-only accounting would understate it.

C Corpus construction

The episodic corpus is generated by `src/energymem/eval/corpus.py`, deterministically from a seed. It contains 122 storylines drawn from 11 topic families (property, travel, health, gardening, vehicle, employment, study, finance, pets, home repair, social), each a chronological sequence of 3–8 episodes, interleaved in time so that the store is not grouped by topic. Multi-target questions are attached to storylines with the gold episode indices recorded at generation time.

Two deliberate difficulty devices:

Near-miss distractors. Each storyline has a planted counterpart in the same topic family that shares vocabulary but not referent — the apartment-hunting thread and the cottage thread both discuss commute times and deposits. These exist so that lexical overlap is an active hazard rather than an accident.

Compositional filler. 900 additional episodes generated from a subject/verb/time template grammar, which inflate the candidate set without introducing new storylines.

The X16 failure cases, in full. These are the clearest evidence in the paper that succession is not similarity, and they are real retrievals from the run, not illustrations. Every returned episode is a reasonable answer to "what is most similar?" and a wrong answer to "what happened next?"

Table 13: Cosine's top-1 for three one-hop association probes.

Cue	Gold (next episode)	Cosine returned
"It turns out to be a forty minute drive from any shop."	"Took most of the food with us and it was fine."	"Decided against Warren Street because the commute would be forty minutes each way." *(matched on "forty minute"; a flat-hunting episode)*
"Planted a crab apple at the end of the bed in October."	"First proper crop this autumn, so the theory held."	"Signed up for the weekly vegetable box." *(matched on growing/vegetables; different storyline)*
"Considering Iceland in February for the northern lights."	"Read that the ring road is often closed that time of year."	"Looking at the fjords in summer rather than winter." *(matched on Nordic travel; the Norway thread)*

Limitation, restated here because this is where a reviewer will look. This corpus is ours. It is generated, not collected, and §8 treats its external validity as unestablished. Appendix E reports the same probe design on a public benchmark built by another group, which is the only evidence that bears on whether the corpus was written to suit the answer.

D The experiment ledger

§7 makes claims about our own process—that errors were caught, that results were overturned by reimplementation, that nothing was quietly rewritten. Those claims are not checkable without the record, so the record ships with the paper.

`docs/10_EXPERIMENT_LEDGER.md` is append-only. Each entry carries an identifier, a date, the question asked, the result, and what the result changed. **An entry is never edited after its result is in.** When a later run overturns an earlier one, that is a new entry pointing back at the old, and the old entry stays as written—so X19 still records the +0.185 two-hop figure in its original wording, and X26b is the entry that withdraws it. A reader can therefore reconstruct not only what we believe now but what we believed then and why we stopped.

The ledger currently runs X1–X26c across five phases: synthetic validation (X1–X6), multi-answer readout (X7–X9), real embeddings (X10–X15), the world-model turn (X16–X19), the mode-averaging gate (X23–X25b), and external validity (X26–X26c).

One gap, stated rather than papered over. Three studies predate the ledger convention and carry no X identifier: the image same-class study, the anchored LoCoMo sweep, and multi-scale β. They are cited in this paper by script name instead. We have deliberately not back-numbered them, because inserting entries into an append-only record retrospectively is precisely the manipulation the record exists to prevent. The ledger preamble says the same. The audit that reinterpreted the first of the three *does* carry an identifier (X113) and a pre-registration committed before its driver existed, which is why the two can be read against each other.

E LoCoMo association probes

Task. Cue is a conversational turn; the gold target is the turn that followed it in the same session. This is the same probe design as the episodic corpus (Appendix C), applied to a benchmark we did not build.

Candidate set. The *full pooled bank* of all 5,882 turns from all ten conversations. A prediction must beat distractors from nine unrelated conversations as well as from its own, which makes the task strictly harder than per-conversation retrieval and is why absolute scores are lower here than on the episodic corpus (cosine Recall@1 0.064 vs. 0.105).

Namespacing. Dialogue identifiers are namespaced by `sample_id`. Without this, 5,882 turns carry only 1,033 distinct identifiers and a turn from an unrelated conversation satisfies a gold identifier and scores as a hit — the bug that invalidated an earlier capacity sweep of ours (§5.2).

Split unit. The *session* (272 of them across ten conversations), not the conversation and not the turn. A session is a coherent conversational episode; transitions from a session the model was fitted on are never evaluated. Four-fold rotation places every session in the test set exactly once, with the validation fold disjoint from both. The world model is fitted on ≈2,800 train-session transitions per fold, and both (k_P, τ) and the readout's (K, β, α) are selected on that fold's validation sessions.

Probe subsampling. Sessions vary in length, so at most 25 probes per session enter the evaluation, sampled with a fixed seed. This keeps folds balanced and costs nothing statistically: the bootstrap resamples sessions, so the interval is driven by the 272 clusters rather than by the probe count.

Full results. Table 14. Note the asymmetry that bounds the claim: at one hop the effect is large and significant on both metrics; at two hops it is null at Recall@1 and carries only a residual +0.012 at Recall@5, an order of magnitude smaller.

The mixture readout, on public data. §5.6 reports that preserving the predictor's modes loses to averaging them on our corpus. It replicates here in sign and magnitude: the mixture beats cosine (+0.011) and loses decisively to averaging (0.075 vs. 0.107, a gap of −0.032). Invariant I4 is falsified on both corpora.

F Statements

Code and data availability. All code, the append-only experiment ledger, and every results file behind every table and figure are released under an open licence at the repository accompanying this paper. Neither evaluation corpus is redistributed: both are public and are fetched by the two commands in Appendix G. The third corpus is generated deterministically from a seed by code in the repository rather than stored, so the exact queries are reproducible without a data file.

Reproducibility. Every table and figure with a results file behind it is regenerated by one command (Appendix G), and the generator prints on every run any table it could not regenerate. Proposition 1 is asserted in continuous integration and writes its agreement counts to a results file.

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Ethics. This work uses two public benchmark corpora and a synthetic corpus of the authors' own construction. It involves no human subjects, no personal data, and no deployment. The agent-memory setting it studies is dual-use only in the ordinary sense that better retrieval improves any system built on it; we identify no specific harm pathway beyond that.

G Reproducibility

Environment. All results were produced on a single machine: Apple M5 Max, 128 GB RAM, macOS 26.5.1 (arm64). Python 3.13.2, NumPy 2.5.2, PyTorch 2.14.0, torchvision 0.29.1, sentence-transformers 6.0.1. No GPU-specific code paths; every experiment runs on CPU. Encoders are frozen pretrained checkpoints (`all-mpnet-base-v2`; `ResNet18_Weights.IMAGENET1K_V1`) and are never fine-tuned.

Data. Neither corpus is redistributed. LoCoMo (2.8 MB) and Imagenette-160 (99 MB) are fetched by the two `curl` commands in the repository README. The episodic corpus is *generated*, not stored: it is produced deterministically from a seed by `src/energymem/eval/corpus.py`, so the exact queries are reproducible without shipping a data file.

Regenerating the paper. Every table and figure that has a results file behind it is produced by one command, and tables are written to `paper/generated/` and `\input`ed rather than typed:

```
make setup
pytest -q                                    # Prop. 1 gate; writes reduction_check.json
python experiments/x23_mode_diagnostics.py   # X23–X25
python experiments/locomo_assoc.py           # X26
python scripts/exp_locomo_anchored.py        # Table 2
python scripts/confirm_holdout.py            # Table 1
python scripts/exp_multibeta_scales.py       # Table 5
python experiments/x113_class_vote.py        # X113; Tables 1 and 6
python experiments/x101_msc.py               # X201; rows (iv) and (d)
python experiments/x10x_corpora.py           # X200, X202–X205; Tables 6 and 7
python experiments/x102_checks.py            # X202c robustness checks
python experiments/x11x_modalities.py        # X210–X212; Table 7
python scripts/make_tables.py                # JSON -> generated tables + figures
cd paper && tectonic -X compile main.tex     # or: latexmk -pdf main
```

`make_tables.py` prints, on every run, the list of tables it could *not* generate from a results file. We regard that output as part of the reproducibility statement: it names precisely which numbers in the paper are still typed rather than derived, so the gap is visible to a reader instead of assumed closed.

Known reproduction frictions. Three, all environmental rather than scientific, recorded because they cost us time. Python on this machine ships without a CA bundle, so `urllib`-based downloads (including `torch.hub` weight fetches) fail with a certificate error; export `SSL_CERT_FILE` from `certifi`, or pre-fetch the checkpoint with `curl`. `torchvision` is an optional dependency not installed by `make setup`; the two image experiments require it. And the machine the final draft was built on carries `tectonic` but neither `pdflatex` nor `latexmk`, so `scripts/rebuild_paper.sh`, which assumes the latter, does not run there; the `tectonic` invocation above is what produced this PDF.

Every number currently reported in this paper is reproducible. Two drivers were lost before they were committed — the one behind Table 4 row (i) and the one behind the parametric-predictor result in §5.3. Both have been rewritten — `x17b_single_split.py` and `x17_parametric.py` under `experiments/` — because the modules themselves survived and only the drivers were missing. No figure in this paper now depends on code nobody can run.

Rewriting them was not free of consequence, and the consequence is reported rather than buried. The parametric result reproduced more starkly than recorded (fitted Recall@1 1.000 and unseen 0.000 at rank 32, against the original 0.784 and 0.008) and the paper cites the re-run. Row (i) did *not* reproduce: the re-run recovers the cosine baseline exactly but gives a non-significant +0.024 at n=123 where the original recorded a significant +0.049 at n=37. The table reports the re-run and marks it n.s.; §5.3 says why the claim does not depend on it. This is the sixth entry in the error taxonomy of §7 and the sharpest form of its argument: **an uncommitted script is not a result.**

Table 11: Seven protocol errors and what each was worth. "Reported" is the number the error produced; "corrected" is the number after the fix. Every one would have been publishable.

Error	**Reported**	**Corrected**	**Direction and why**
Hyperparameter not dimensionless	0.016	0.277	**False negative.** At d=768, a nominal σ=0.35 perturbation applied without a $1/\sqrt{d}$ factor has norm 9.7 against a unit cue, giving $\cos(q, q+\epsilon)$=0.124: the cue is destroyed. The method appeared not to work.
Pseudo-replicated bootstrap	p=0.0005	p=0.076	**False positive.** 185 rows were 5 correlated seeds over the same 37 questions. Resampling rows treats them as independent; resampling questions is the correct cluster. A 150× difference in p.
Test-set hyperparameter selection	+0.042	+0.004 (p=0.43)	**False positive.** β was swept on the questions it was then scored on. Two corrections—more questions, then a storyline-disjoint split—took the effect to zero in stages (+0.042 → +0.027 → +0.004).
Asymmetric candidate exclusion	0.000 exactly	—	**False negative, disguised as a finding.** The cue was excluded from the candidate set for the baseline but not for the method under test. An exact zero alongside an unchanged Recall@5 looked like a structural result.
Unmatched step size	−0.115	−0.062	**Inflated effect.** Comparing predictor variants at α=1 against an average that implicitly applies $\alpha\approx$0.34 confounds the comparison with step size. Roughly half the apparent effect was the confound.
Unmatched baseline	+0.014 [+0.006, +0.021]	−0.036 [-0.043, -0.028]	**False positive, and the only one no protocol check caught.** The image comparison was a 1-nearest-neighbour classifier against a soft-k-nearest-neighbour one, scored on a ten-way *class* metric. Split hygiene, pre-registration and cluster bootstraps were all in place and all irrelevant: the baseline was weaker than the method for a reason unrelated to the mechanism under test (X113).
Uncommitted script	+0.049 (p=0.032)	+0.024 (p=0.24)	**Unresolvable.** A significant result at n=37 whose driver was never committed. Re-running the same split recovers the cosine baseline exactly but yields n=123 probes and no significant effect. With the original code gone the discrepancy cannot be diagnosed — only reported.

Table 12: Every grid searched, where it was searched, and where the number was reported. "Budget" is the number of configurations evaluated. Baselines receive a budget at least as large as the method they are compared against, which is the commitment §4 makes.

Experiment	Parameter grid	Budget	Selected on	Reported on
Image, same-class	$\beta \in \{12, \ldots, 512\}$, 14 values	14	tune half (images)	held-out half
	cosine: none	—	—	held-out half
	k-NN vote: $k \times \tau$, 6×5	30	tune half (images)	held-out half
Image, same-image (X113B)	$\beta \in \{24, \ldots, 80\}$, 10 values	10	tune half (bank images)	held-out half
Multi-β	scale sets, 6 conditions	6	tune half	held-out half
LoCoMo anchored	`anchor` $\times$ pp_0, 8 conditions	8	—	all 1,977 questions
	BM25: k_1, b grid	9	dev questions	test questions
	hybrid RRF: k constant	5	dev questions	test questions
Readout variants	$\beta \in \{16, 32, 64, 128\}$	4	dev storylines	test storylines
Association (ours)	$k_P \in \{5, 10, 20\}$, $\tau \in \{2, 5, 10\}$	9	val storylines, per fold	test storylines
	mixture K, β, α	48	val storylines, per fold	test storylines
Association (LoCoMo)	k_P, τ as above	9	val sessions, per fold	test sessions
	mixture K, β, α	18	val sessions, per fold	test sessions
Association (MSC, X201)	LoCoMo's grid, unchanged	9	val sessions, per fold	test sessions
Six corpora (X200–X204)	expansion: $K_0 \in \{0, 1, 2, 3, 5\} \times \beta \in \{4, 16, 64\} \times$ `with_query`	30	9 of 10 groups, per fold	held-out group
	averaged expansion: same grid	30	9 of 10 groups, per fold	held-out group
	Rocchio: $K_0 \times \gamma \in \{0, 0.25, 0.5, 1\}$	20	9 of 10 groups, per fold	held-out group
	MMR: $\lambda \in \{0.3, \ldots, 0.9\} \times$ `pool` $\in \{20, 50\}$	8	9 of 10 groups, per fold	held-out group
	RRF: $k_{\text{rrf}} \in \{10, 60, 120\}$	3	9 of 10 groups, per fold	held-out group
	BM25, cosine: none	—	—	held-out group
Modalities (X211, X212)	expansion and Rocchio grids as above	30	9 of 10 groups, per fold	held-out group

Table 14: LoCoMo association, all methods and both hops. Hop 1 n=5,233; hop 2 n=5,013; 272 session clusters throughout; intervals are 95% cluster bootstrap. `experiments/locomo_assoc.py`; raw values in `experiments/results/x26_locomo_assoc.json`.

Hop	Method	R@1	Δ R@1	R@5	Δ R@5
1	cosine	0.064	—	0.174	—
	averaged	**0.107**	+0.043 [+0.036, +0.050]	**0.251**	+0.077 [+0.067, +0.087]
	mixture	0.075	+0.011 [+0.006, +0.016]	0.202	+0.028 [+0.022, +0.035]
2	cosine	0.043	—	0.127	—
	averaged	0.041	−0.002 [-0.008, +0.004]	0.139	+0.012 [+0.004, +0.019]
	mixture	0.045	+0.002 [-0.003, +0.006]	0.134	+0.007 [+0.002, +0.012]